ENGLISH FOR WORK / PRODUCTION / IMPROVEMENT / RELIABILITY / HANDOFFS

Manufacturing English

Describe the loss. Test the explanation. Make the handoff clear.



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Eight practical lessons. Eleven extended conversations.
Vocabulary, phrase banks, guided speaking, and explained answers.

Revised 2026 edition / Upper-intermediate to advanced

Your route through the book

01 Production Flow and Daily Management, page 4

Report an output shortfall precisely while separating observed downtime, unconfirmed causes, and different performance measures.

02 Lean, Waste, and Continuous Improvement, page 14

Describe observed waiting, challenge unsupported blame, and define a bounded improvement trial without claiming savings in advance.

03 Defects, Scrap, and Rework, page 27

Reconcile output categories and explain yield, pending recovery, and scrap without double-counting or assuming a rework outcome.

04 Root Cause and Corrective Action, page 37

Challenge a premature root-cause statement, preserve competing hypotheses, and distinguish containment from verified corrective action.

05 Maintenance, Reliability, and Changeover, page 50

Explain schedule conflicts, distinguish work types, and present conditional options without assuming authority to defer maintenance or restart equipment.

06 EHS and Safety Communication, page 61

Raise an ambiguous safety instruction clearly, distinguish shutdown from energy control, and confirm the authorized clarification route.

07 Supplier Quality and Incoming Materials, page 73

Compare measured results against explicit revision limits and resolve the governing requirement before assigning responsibility or release status.

08 Shift Handoffs and Escalation, page 83

Transfer unfinished work with specific identifiers, evidence, ownership, dependencies, and a confirmed next update.

Professional communication notes, page 3

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Suggested study range: B2-C1. This is an editorial study range, not a certified proficiency assessment.

Keep the number attached to its meaning.

Manufacturing discussions move quickly between quantities, causes, schedules, and decisions. A missed target may be real even when its cause is unknown. A held lot may contain uninspected material. A named action may still be unfinished. These cases develop the language for keeping those distinctions visible under the everyday pressure to restore output and meet commitments.

State the measure and its basis

Give the unit, time period, numerator, and comparison. Output attainment, availability, and first-pass yield are different measures; a familiar percentage should not be relabeled without the required data.
"We achieved 84% of the output target; that is not an OEE result."

Observe before assigning a cause

A timeline or repeated symptom can guide investigation without proving a root cause. State what changed, what was observed, and which competing explanations remain open.
"The defect followed the supplier change, but the process settings changed too."

Separate material states

Scrapped, awaiting rework, on hold, accepted, and released have different consequences. Use quantities for each state and avoid counting expected recovery as finished good output.
"Sixty units await rework; they are not yet accepted units."

Make the handoff closed-loop

A sent message does not prove it was understood. Identify the open item, evidence, responsible person, and next checkpoint, then confirm receipt and understanding without implying completion.
"Two checks remain open; please confirm their identifiers and the next update."

Professional boundaries

Original fictional language practice, not equipment-operation, maintenance, engineering, or safety instructions. Figures, organizations, and workplace cases are invented. Actual work must follow applicable law, approved site procedures, trained personnel, and authorized decisions. Classroom safety discussions do not authorize isolation, servicing, or restart.

A Read the situation

LESSON 01 / PRODUCTION FLOW AND DAILY MANAGEMENT

Report an output shortfall precisely while separating observed downtime, unconfirmed causes, and different performance measures.

Eight hundred forty is not the whole story

Production lead Mei and operations manager Daniel review yesterday's line output: 840 completed units against a 1,000-unit target. The log records 40 minutes of downtime, but the cause and its contribution to the shortfall are still being checked. The briefing supplies no planned production time, ideal cycle time, or quality breakdown. A dashboard labels 84% as overall equipment effectiveness. Mei and Daniel must correct the metric and give a useful update without claiming the downtime explains every missing unit.

People: Mei, Production lead; Daniel, Operations manager.

Read the record

Choose one answer per question using the facts above.

1. What was the output shortfall?

A. 40 units
B. 840 units
C. 1,840 units
D. 160 units

2. What does 84% represent here?

A. Output as a share of the stated target
B. A supplied OEE calculation
C. A verified availability percentage
D. A first-pass yield result

3. What is known about the downtime?

A. It was caused by operator error.
B. It proves the line had no quality losses.
C. Forty minutes were logged; its cause and contribution remain under review.
D. It definitely explains all 160 missing units.

Answers and explanations, page 108

B Vocabulary in use

CORE TERMS

production target
The planned output quantity for a defined period.

set a production target

actual output
The quantity actually produced in the stated period.

report actual output

target attainment
Actual output expressed relative to the stated target.

calculate target attainment

shortfall
The amount by which a result falls below its target.

quantify the shortfall

throughput
The quantity passing through a process per unit of time.

measure throughput

cycle time
The elapsed time required for a defined process cycle.

measure cycle time

takt time
Available production time divided by required demand for the same period.

calculate takt time

lead time
The elapsed time from a defined request or start to completion.

reduce lead time

work in process
Material that has entered production but is not yet finished.

control work in process

bottleneck
The step that limits the output of the overall process.

identify the bottleneck

constraint
A limiting factor on the system's performance.

manage the constraint

line balance
Distribution of work across production steps.

improve line balance

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

production target PRODUCTION TARGET

set a production target

Keep the meaning precise: The planned output quantity for a defined period.

cycle time CYCLE TIME

measure cycle time

Keep the meaning precise: The elapsed time required for a defined process cycle.

B Precision and professional shorthand

EXTEND YOUR RANGE

downtime

Time during which equipment or a process is not producing as defined.

log downtime

planned stop

A scheduled interruption included in the operating plan.

distinguish planned stops

unplanned stop

An interruption not included in the intended operating schedule.

investigate unplanned stops

availability

The share of planned production time in which equipment is running.

assess equipment availability

performance loss

Output-speed loss relative to the defined ideal running rate.

analyze performance losses

quality loss

Output that does not meet the defined good-product criteria.

quantify quality losses

overall equipment effectiveness

Availability multiplied by performance and quality, with defined factors; abbreviated OEE.

calculate overall equipment effectiveness

daily management

Routine review of performance, problems, and follow-up actions.

support daily management

tier meeting

A short escalation meeting at a defined organizational level.

escalate through a tier meeting

visual board

A shared display of current measures, status, and actions.

update the visual board

loss attribution

Assignment of a measured loss to supported contributing factors.

verify loss attribution

recovery plan

A defined proposal for addressing a performance or schedule gap.

review the recovery plan

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

line balance LINE BALANCE

improve line balance

Keep the meaning precise: Distribution of work across production steps.

performance loss PERFORMANCE LOSS

analyze performance losses

Keep the meaning precise: Output-speed loss relative to the defined ideal running rate.

C Phrases that move the work forward

Lead with the result

We completed 840 units against a target of 1,000.

Quantify the gap

The shortfall is 160 units, or 16% of target.

Name the measure

Eighty-four percent is output target attainment.

Correct the label

The supplied figures do not support an OEE calculation.

State the logged loss

The log records 40 minutes of downtime.

Keep cause open

The reason for that downtime is still being checked.

Limit attribution

We have not established how much of the shortfall the stop explains.

Ask for the time basis

What planned production time was used in this calculation?

Distinguish rate from quantity

Throughput needs a quantity and a time period.

Check the constraint

We have not yet confirmed which step limited total output.

Request the breakdown

Separate speed, availability, and quality information.

Avoid blame

The current log does not establish an operator cause.

Make the update useful

Report the known gap and the next evidence check separately.

Bound the recovery claim

A recovery proposal is not a guarantee that the missing output will be recovered.

Assign the follow-up

Mei will reconcile the log and report the supported loss breakdown.

Close the board update

Keep the metric definition beside the percentage.

Precision in real exchanges

Purpose: Correct the label
Useful wording: The supplied figures do not support an OEE calculation.

Purpose: State the logged loss
Useful wording: The log records 40 minutes of downtime.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which board entry is accurate?

A. OEE 84%, with all losses explained.

B. Availability 84% and first-pass yield 100%.

C. Forty missing units because downtime was 40 minutes.

D. 840 units completed; 84% of target; 160-unit shortfall.

2. Why is OEE not established?

A. Downtime alone supplies every OEE input.

B. Planned time, ideal cycle time, and quality information are not supplied.

C. OEE is always identical to target attainment.

D. A target of 1,000 automatically defines ideal cycle time.

3. A colleague needs the term for this meaning: The amount by which a result falls below its target. Which term fits?

A. work in process

B. unplanned stop

C. tier meeting

D. shortfall

D Choose the response that fits / continued

4. Rebuild the exchange from "Eight hundred forty is not the whole story". Daniel: "The stop log records forty minutes. Does the team believe that interruption accounts for all the missing units, or is the analysis still open?" Mei: [missing response] Daniel: "We would need to know the expected production rate and whether other losses occurred. The board currently makes a single interruption look like a complete explanation." Which response belongs in the gap?

A. The shortfall is one hundred sixty units, or sixteen percent of target. That is a quantity gap, not a duration and not an explanation of the cause.

B. I will update the visual board accordingly. It will show the output gap and logged stop separately, with me responsible for reconciling the records and reporting the supported breakdown.

C. The downtime is logged, but its cause and contribution remain under review. We should not convert forty minutes into one hundred sixty lost units without the relevant basis.

D. Exactly. Loss attribution needs supporting data. I will reconcile the timing, output, and other relevant records rather than attach the entire gap to the most visible event.

Answers and explanations, page 108

One useful distinction

840 divided by 1,000 is 84% target attainment. The shortfall is 160 units, or 16% of target. Neither result supplies OEE, availability, or first-pass yield. Those measures require their own defined inputs and calculation bases.

E Conversation

Eight hundred forty is not the whole story

Eight hundred forty is not the whole story

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

target attainment / availability / recovery plan / performance loss / visual board / shortfall / Quality loss / Loss attribution / downtime / bottleneck

01 **Daniel:** Yesterday's board shows eight hundred forty units against a thousand. It labels eighty-four percent as OEE, but I cannot see the inputs behind that label.

02 **Mei:** The figure is [1] _____. We divided completed output by the stated target; we did not calculate the availability, performance, and quality factors.

03 **Daniel:** Please change the heading before the morning meeting. What is the gap in units, and what can we actually say about the stop?

04 **Mei:** The [2] _____ is one hundred sixty units, or sixteen percent of target. That is a quantity gap, not a duration and not an explanation of the cause.

05 **Daniel:** The stop log records forty minutes. Does the team believe that interruption accounts for all the missing units, or is the analysis still open?

06 **Mei:** The [3] _____ is logged, but its cause and contribution remain under review. We should not convert forty minutes into one hundred sixty lost units without the relevant basis.

07 **Daniel:** We would need to know the expected production rate and whether other losses occurred. The board currently makes a single interruption look like a complete explanation.

08 **Mei:** Exactly. [4] _____ needs supporting data. I will reconcile the timing, output, and other relevant records rather than attach the entire gap to the most visible event.

09 **Daniel:** Which inputs do you need for the equipment measure? Give me the missing fields so I can ask the line team for them.

10 **Mei:** Start with the basis for [5] _____. We need the planned production time and the relevant running time, not merely the output target and one stop duration.

Conversation continues on the next page.

Eight hundred forty is not the whole story

CONVERSATION / CONTINUED

11 **Daniel:** The speed side also needs a clear reference. Actual output can fall short even while the equipment is running, so time and quantity should not be merged.

12 **Mei:** Yes. The [6] _____ assessment needs the defined ideal rate and actual running performance. The supplied briefing does not provide that basis, so the board cannot claim a completed result.

13 **Daniel:** And the completed-unit count does not tell us how many met the quality definition. We should not silently assume that every produced unit was good.

14 **Mei:** Correct. [7] _____ needs the appropriate output breakdown. A production count by itself cannot establish that no defects or other quality losses occurred.

15 **Daniel:** Someone has suggested the packing station is the limiting step. That may be worth checking, but I have not seen evidence showing it constrained the whole line.

16 **Mei:** Keep [8] _____ as an investigation question. A busy station or visible queue does not automatically identify the system's actual limiting step.

17 **Daniel:** For today's meeting, I need a concise update: result, known interruption, open questions, and responsibility for the next check. We can discuss recovery after that evidence is clearer.

18 **Mei:** I will update the [9] _____ accordingly. It will show the output gap and logged stop separately, with me responsible for reconciling the records and reporting the supported breakdown.

19 **Daniel:** Good. We can acknowledge the missed target without guessing why it happened. Any proposal to recover output should make its assumptions and constraints clear.

20 **Mei:** Agreed. The [10] _____ will remain a proposal until assessed and authorized. First, the board must use the right measure and keep the unfinished investigation visible.

CHECK THE GAPS

1. target attainment / **2.** shortfall / **3.** downtime / **4.** Loss attribution / **5.** availability / **6.** performance loss / **7.** Quality loss / **8.** bottleneck / **9.** visual board / **10.** recovery plan

Why these answers fit, page 110

F Check accuracy and scope

Eight hundred forty is not the whole story

Return to the case

Production lead Mei and operations manager Daniel review yesterday's line output: 840 completed units against a 1,000-unit target. The log records 40 minutes of downtime, but the cause and its contribution to the shortfall are still being checked. The briefing supplies no planned production time, ideal cycle time, or quality breakdown. A dashboard labels 84% as overall equipment effectiveness. Mei and Daniel must correct the metric and give a useful update without claiming the downtime explains every missing unit.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. A manager converts the 40-minute stop into the full 160-unit shortfall. What is missing from that attribution?

A. A supported running-rate and loss breakdown linking the stop to the output gap

B. The subtraction of 840 completed units from the 1,000-unit target

C. Confirmation that 40 is smaller than the target of 1,000

D. A relabeling of target attainment as availability

2. If a second shift produces 900 against a target of 1,000, what is its attainment?

A. 90%, six percentage points above 84%

B. 90%, exactly six percent above 84%

C. 10%, because 100 units are missing

D. 900%, because 900 units were produced

Answers and explanations, page 111

Listen for the distinction

Swap roles. Keep the forty-minute stop separate from its unconfirmed cause and output effect; do not label 84% as OEE.

A precise update can still be brief

Start with the result, then the confirmed loss information and the open cause. This lets a busy manager distinguish the size of the gap from the explanation. Avoid filling a short status slot with a confident causal claim simply because the investigation is unfinished.

F Say it / transfer to a new case

Compare the right percentages

A line produces 450 units against a target of 500. It logs 20 minutes of downtime. No cause or OEE inputs are supplied.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

downtime / ninety / fifty / unconfirmed

01 **Lead:** Target attainment is [1] _____ percent.

02 **Planner:** The output shortfall is [2] _____ units.

03 **Lead:** Twenty minutes describes recorded [3] _____.

04 **Planner:** The cause remains [4] _____.

CHECK THE GAPS

1. ninety / **2.** fifty / **3.** downtime / **4.** unconfirmed

Why these answers fit, page 111

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Eight hundred forty is not the whole story". Daniel: "And the completed-unit count does not tell us how many met the quality definition. We should not silently assume that every produced unit was good." Mei: [missing response] Daniel: "Someone has suggested the packing station is the limiting step. That may be worth checking, but I have not seen evidence showing it constrained the whole line." Which response belongs in the gap?

A. The figure is target attainment. We divided completed output by the stated target; we did not calculate the availability, performance, and quality factors.

B. Keep bottleneck as an investigation question. A busy station or visible queue does not automatically identify the system's actual limiting step.

C. Start with the basis for availability. We need the planned production time and the relevant running time, not merely the output target and one stop duration.

D. Correct. Quality loss needs the appropriate output breakdown. A production count by itself cannot establish that no defects or other quality losses occurred.

Answers and explanations, page 112

A Read the situation

LESSON 02 / LEAN, WASTE, AND CONTINUOUS IMPROVEMENT

Describe observed waiting, challenge unsupported blame, and define a bounded improvement trial without claiming savings in advance.

> **The queue is for a shared tool**
>
> Improvement facilitator Ana and supervisor Marcus observe a one-hour assembly session. Eight waiting episodes for a shared tool total 24 operator-minutes; some episodes overlap, so this is not 24 minutes of line downtime. The supervisor calls the operators slow. The team proposes a one-bench point-of-use tool trial, but no trial result exists. Ana and Marcus must distinguish waiting from working pace, explain the time basis, and keep safety and quality checks unchanged in the proposal.

People: Ana, Improvement facilitator; Marcus, Assembly supervisor.

Read the record

Choose one answer per question using the facts above.

1. What was observed?

A. Twenty-four minutes of confirmed line downtime
B. A proven slow working pace for every operator
C. A successful completed tool trial
D. Eight tool-waiting episodes totaling 24 operator-minutes

2. Why is 24 not necessarily line downtime?

A. The line was proved to run without any delay.
B. Some waiting episodes overlap, and the measure is operator-minutes.
C. Operator-minutes never include waiting.
D. Eight episodes always equal eight hours.

3. What is the proposed change?

A. A guaranteed saving already verified
B. A one-bench point-of-use tool trial with safety and quality checks unchanged
C. Immediate removal of all inspection steps
D. A completed factory-wide rollout

Answers and explanations, page 113

B Vocabulary in use

CORE TERMS

lean manufacturing
An approach to improving customer value by reducing waste in the process.

apply lean manufacturing

waste
Resource use that does not add the defined value or support a necessary requirement.

identify process waste

waiting
Time when work cannot proceed because a needed input is unavailable.

reduce waiting time

motion
Movement of people within a task or process.

analyze unnecessary motion

transportation
Movement of material between locations or steps.

reduce unnecessary transportation

overproduction
Producing earlier or in greater quantity than needed.

prevent overproduction

overprocessing
Performing more work than the relevant requirement needs.

identify overprocessing

excess inventory
Material held beyond the justified operating need.

reduce excess inventory

kaizen
Focused, ongoing improvement of work processes.

run a kaizen event

gemba
The actual place where the work occurs.

observe at the gemba

value stream
The connected activities and information involved in delivering a product.

map the value stream

current-state map
A representation of how the process currently works.

create a current-state map

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

lean manufacturing LEAN MANUFACTURING

apply lean manufacturing

Keep the meaning precise: An approach to improving customer value by reducing waste in the process.

overproduction OVERPRODUCTION

prevent overproduction

Keep the meaning precise: Producing earlier or in greater quantity than needed.

B Precision and professional shorthand

EXTEND YOUR RANGE

future-state map

A proposed representation of an improved process.

develop a future-state map

5S

Workplace organization through sort, set in order, shine, standardize, and sustain.

sustain 5S practices

pull system

Replenishment or production triggered by downstream need.

design a pull system

work-cell layout

The arrangement of people, equipment, and material in a production cell.

assess the work-cell layout

baseline observation

A measurement of the current process before a proposed change.

record a baseline observation

countermeasure

An action intended to address an identified problem.

test a countermeasure

point-of-use storage

Keeping needed items near where they are used.

trial point-of-use storage

standard work

The agreed method and conditions for performing a process.

update standard work

kanban

A signal authorizing defined replenishment or production in a pull system.

use a kanban signal

operator-minute

One minute of one operator's time, accumulated across people if specified.

report operator-minutes

pilot trial

A limited test of a proposed change before broader adoption.

define a pilot trial

sustainment

Maintaining an improvement over time through defined follow-up.

plan sustainment

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

current-state map CURRENT-STATE MAP

create a current-state map

Keep the meaning precise: A representation of how the process currently works.

pull system PULL SYSTEM

design a pull system

Keep the meaning precise: Replenishment or production triggered by downstream need.

C Phrases that move the work forward

Describe the delay

Operators waited for the shared tool during eight observed episodes.

Name the time basis

The total is 24 operator-minutes, not 24 minutes of line downtime.

Separate pace from access

Waiting for a tool does not by itself show slow working pace.

Challenge the attribution

The observation does not establish that operator speed caused the delay.

Invite practical detail

Ask the operators where the tool is needed and what constrains access.

Define the current state

Record the present locations, demand, and waiting pattern.

State the proposal

Trial point-of-use storage at one bench.

Bound the claim

No improvement result is available yet.

Preserve controls

Required safety and quality checks remain unchanged.

Compare consistently

Use the same time basis and comparable task conditions.

Check for displaced problems

Confirm that the change does not create a shortage at another bench.

Separate a trial from rollout

A one-bench test is not a factory-wide implementation.

Avoid prebooked savings

Do not report proposed time savings as already achieved.

Make the review specific

Review the waiting observations and any new difficulties after the trial.

Retain operator input

Include the people who use the tool in the practical review.

Plan the follow-through

Agree how an accepted improvement would be maintained.

Precision in real exchanges

Purpose: Challenge the attribution
Useful wording: The observation does not establish that operator speed caused the delay.

Purpose: Invite practical detail
Useful wording: Ask the operators where the tool is needed and what constrains access.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which observation is defensible?

A. The entire line stopped for 24 minutes.

B. The proposed trial saved 24 minutes already.

C. Eight tool-waiting episodes totaled 24 operator-minutes during the hour.

D. Operators were lazy for exactly 40% of the shift.

2. What is the average time per recorded episode?

A. Twenty-four hours

B. Forty percent of every operator's shift

C. Three operator-minutes

D. Eight operator-minutes

3. A colleague needs the term for this meaning: Movement of people within a task or process. Which term fits?

A. 5S

B. baseline observation

C. motion

D. kaizen

D Choose the response that fits / continued

4. Rebuild the exchange from "The queue is for a shared tool". Marcus: "I will put the one-hour observation period beside that figure. We did not watch a full shift, so I cannot call it a shift total." Ana: [missing response] Marcus: "The operators suggested a tool location beside the bench. It sounds practical, but we need to understand whether other stations depend on that same shared tool." Which response belongs in the gap?

A. Correct it to proposed countermeasure. We can explain what the change is intended to address without booking the hoped-for benefit as an achieved result.

B. The measure is operator-minute. The total accumulates individual waiting time, so overlapping episodes must not be presented as the same amount of elapsed whole-line downtime.

C. Exactly. This is our baseline observation. Record the task conditions and tool demand so the later comparison is meaningful rather than just another isolated total.

D. Review the broader value stream. Local gains matter, but a change that simply transfers waiting to another step may not improve delivery to the customer.

Answers and explanations, page 113

> **One useful distinction**
>
> Twenty-four operator-minutes across eight episodes averages three operator-minutes per episode. Overlapping waits mean that total is not automatically elapsed line downtime. State the observation period and time basis when reporting the result.

E Conversation

The queue is for a shared tool

The queue is for a shared tool

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

pilot trial / waiting / baseline observation / point-of-use storage / motion / current-state map / countermeasure / value stream / operator-minute / sustainment

01 **Marcus:** We watched the bench for an hour and recorded eight delays. The team says the operators are slow, but each delay seemed to involve waiting for the shared tool.

02 **Ana:** Then name the observed [1] _____. The tool was unavailable at those moments; that does not establish that the operators worked slowly when they had what they needed.

03 **Marcus:** The sheet adds up to twenty-four minutes. I was going to call that twenty-four minutes of line downtime, although two operators sometimes waited at the same time.

04 **Ana:** The measure is [2] _____. The total accumulates individual waiting time, so overlapping episodes must not be presented as the same amount of elapsed whole-line downtime.

05 **Marcus:** I will put the one-hour observation period beside that figure. We did not watch a full shift, so I cannot call it a shift total.

06 **Ana:** Exactly. This is our [3] _____. Record the task conditions and tool demand so the later comparison is meaningful rather than just another isolated total.

E Conversation / continued

07 **Marcus:** The operators suggested a tool location beside the bench. It sounds practical, but we need to understand whether other stations depend on that same shared tool.

08 **Ana:** Create the [4] _____ with their input. Show the tool's movements and users, not just the bench we observed.

09 **Marcus:** That would also help us separate people walking to retrieve the tool from material being moved between work areas. Both happen, but they are different kinds of movement.

10 **Ana:** Yes. Unnecessary [5] _____ concerns the people performing the task. We should describe the actual movement rather than use a broad waste label that hides what needs to change.

Conversation continues on the next page.

The queue is for a shared tool

CONVERSATION / CONTINUED

11 **Marcus:** For the trial, the suggestion is to keep an appropriate tool at this one bench. We have not decided that every work area needs a new tool arrangement.

12 **Ana:** That is a proposal for [6] _____. Check the practical requirements and shared demand before assuming that moving the tool solves the problem for the whole process.

13 **Marcus:** We can start with one bench and compare the waiting pattern. Required inspection and safety checks should stay in place, so the trial does not gain time by removing controls.

14 **Ana:** Make that explicit in the [7] _____ description. Use comparable task conditions and the same measure, and record any new difficulties rather than reporting only the favorable observations.

15 **Marcus:** Hold on, the slide already says twenty-four minutes saved. Nobody has run the trial. Can we change that before it reaches the review meeting?

16 **Ana:** Correct it to proposed [8] _____. We can explain what the change is intended to address without booking the hoped-for benefit as an achieved result.

17 **Marcus:** After the trial, we should ask whether a reduction here created a tool shortage elsewhere. A faster bench would not necessarily mean a better overall flow.

18 **Ana:** Review the broader [9] _____. Local gains matter, but a change that simply transfers waiting to another step may not improve delivery to the customer.

19 **Marcus:** If the trial supports the change, the team still needs an approved method and follow-up. Otherwise the tool could gradually return to the old shared location.

20 **Ana:** That is [10] _____. Update the agreed method through the proper process, clarify ownership, and monitor whether the improvement persists without weakening the required controls.

CHECK THE GAPS

1. waiting / **2.** operator-minute / **3.** baseline observation / **4.** current-state map / **5.** motion / **6.** point-of-use storage / **7.** pilot trial / **8.** countermeasure / **9.** value stream / **10.** sustainment

Why these answers fit, page 114

F Check accuracy and scope

The queue is for a shared tool

Return to the case

Improvement facilitator Ana and supervisor Marcus observe a one-hour assembly session. Eight waiting episodes for a shared tool total 24 operator-minutes; some episodes overlap, so this is not 24 minutes of line downtime. The supervisor calls the operators slow. The team proposes a one-bench point-of-use tool trial, but no trial result exists. Ana and Marcus must distinguish waiting from working pace, explain the time basis, and keep safety and quality checks unchanged in the proposal.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which proposal has suitable boundaries?

A. Remove checks to guarantee a faster result.

B. Roll out everywhere before reviewing the local effect.

C. Declare all waits eliminated before the trial.

D. Trial tool placement at one bench, keep required controls, and compare waiting under comparable conditions.

2. Waiting falls at the trial bench while the shared tool is less available elsewhere. Which conclusion still needs evidence?

A. The existence of a proposal to change the tool location

B. Total benefit across the connected process, including any displaced waiting

C. The fact that the observation came from a particular bench

D. The original eight waiting episodes in the supplied baseline

Answers and explanations, page 116

Listen for the distinction

Swap roles. Contrast accumulated people-time with line downtime; keep the one-bench trial proposed and controls unchanged.

Invite the people who do the work

A respectful observation describes the delay and asks how the process creates it. Operators can explain practical constraints that a meeting-room proposal misses. Their involvement is not permission to bypass controls; it helps the team design a realistic, reviewable improvement trial.

F Say it / transfer to a new case

Add people-time, not clock-time

Four overlapping waiting episodes total 12 operator-minutes in a 30-minute observation. A new tool location is proposed, but no trial has run.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

operator-minutes / trial / savings / three

01 **Facilitator:** The accumulated waiting measure is [1] _____.

02 **Supervisor:** Average waiting per episode is [2] _____ minutes.

03 **Facilitator:** The proposed tool location still needs a [3] _____.

04 **Supervisor:** We cannot report achieved [4] _____ yet.

CHECK THE GAPS

1. operator-minutes / **2.** three / **3.** trial / **4.** savings

Why these answers fit, page 116

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "The queue is for a shared tool". Marcus: "We can start with one bench and compare the waiting pattern. Required inspection and safety checks should stay in place, so the trial does not gain time by removing controls." Ana: [missing response] Marcus: "Hold on, the slide already says twenty-four minutes saved. Nobody has run the trial. Can we change that before it reaches the review meeting?" Which response belongs in the gap?

A. That is a proposal for point-of-use storage. Check the practical requirements and shared demand before assuming that moving the tool solves the problem for the whole process.

B. Make that explicit in the pilot trial description. Use comparable task conditions and the same measure, and record any new difficulties rather than reporting only the favorable observations.

C. Then name the observed waiting. The tool was unavailable at those moments; that does not establish that the operators worked slowly when they had what they needed.

D. Yes. Unnecessary motion concerns the people performing the task. We should describe the actual movement rather than use a broad waste label that hides what needs to change.

Answers and explanations, page 117

Review 1 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Production Flow and Daily Management" and "Lean, Waste, and Continuous Improvement". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: Actual output expressed relative to the stated target. Which term fits?

A. lead time
B. planned stop
C. target attainment
D. daily management

2. A colleague needs the term for this meaning: Material held beyond the justified operating need. Which term fits?

A. lean manufacturing
B. future-state map
C. work-cell layout
D. excess inventory

3. Rebuild the exchange from "Eight hundred forty is not the whole story". Daniel: "Which inputs do you need for the equipment measure? Give me the missing fields so I can ask the line team for them." Mei: [missing response] Daniel: "The speed side also needs a clear reference. Actual output can fall short even while the equipment is running, so time and quantity should not be merged." Which response belongs in the gap?

A. Exactly. Loss attribution needs supporting data. I will reconcile the timing, output, and other relevant records rather than attach the entire gap to the most visible event.
B. The shortfall is one hundred sixty units, or sixteen percent of target. That is a quantity gap, not a duration and not an explanation of the cause.
C. Start with the basis for availability. We need the planned production time and the relevant running time, not merely the output target and one stop duration.
D. Agreed. The recovery plan will remain a proposal until assessed and authorized. First, the board must use the right measure and keep the unfinished investigation visible.

Review 1 / Meaning and context / continued

4. Rebuild the exchange from "The queue is for a shared tool". Marcus: "After the trial, we should ask whether a reduction here created a tool shortage elsewhere. A faster bench would not necessarily mean a better overall flow." Ana: [missing response] Marcus: "If the trial supports the change, the team still needs an approved method and follow-up. Otherwise the tool could gradually return to the old shared location." Which response belongs in the gap?

A. Exactly. This is our baseline observation. Record the task conditions and tool demand so the later comparison is meaningful rather than just another isolated total.

B. The measure is operator-minute. The total accumulates individual waiting time, so overlapping episodes must not be presented as the same amount of elapsed whole-line downtime.

C. Review the broader value stream. Local gains matter, but a change that simply transfers waiting to another step may not improve delivery to the customer.

D. Correct it to proposed countermeasure. We can explain what the change is intended to address without booking the hoped-for benefit as an achieved result.

Answers and explanations, page 148

A Read the situation

LESSON 03 / DEFECTS, SCRAP, AND REWORK

Reconcile output categories and explain yield, pending recovery, and scrap without double-counting or assuming a rework outcome.

> **One thousand units, three different states**
>
> Quality analyst Priya and production manager Ellis review 1,000 inspected units from one batch. Exactly 900 passed on the first inspection, 60 are awaiting authorized rework, and 40 have an approved scrap disposition. These categories are mutually exclusive and account for all 1,000 units. No reworked units have yet been accepted. A summary labels all 100 non-first-pass units as scrap and forecasts 960 good units as though they already exist. Priya must correct both the loss and recovery statements.

People: Priya, Quality analyst; Ellis, Production manager.

Read the record

Choose one answer per question using the facts above.

1. What is the first-pass yield?

A. 94%
B. 10%
C. 90%
D. 96%

2. How many units have approved scrap disposition?

A. 100
B. 60
C. 900
D. 40

3. How many reworked units are already accepted?

A. 960
B. None
C. 60
D. 100

Answers and explanations, page 118

B Vocabulary in use

CORE TERMS

defect
A product characteristic that fails an applicable quality requirement.

classify a defect

defective unit
A unit containing one or more defects under the stated definition.

count defective units

nonconforming unit
A unit that does not meet a specified requirement.

identify nonconforming units

scrap
Product designated not to be used for its original intended purpose.

record approved scrap

rework
Action intended to bring nonconforming product into conformity.

perform authorized rework

repair
Action making product acceptable for intended use without necessarily meeting every original requirement.

distinguish repair from rework

first-pass yield
The share of units meeting requirements on their first pass.

calculate first-pass yield

final yield
The share meeting the stated acceptance criteria after the defined processing stages.

state the final-yield basis

rework queue
Units waiting for the approved rework process.

monitor the rework queue

scrap rate
The scrapped quantity divided by the stated reference quantity.

report the scrap rate

disposition
An authorized decision about how identified material will be handled.

approve a disposition

acceptance status
Whether the required acceptance decision has been completed.

verify acceptance status

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

defect DEFECT

classify a defect

Keep the meaning precise: A product characteristic that fails an applicable quality requirement.

repair REPAIR

distinguish repair from rework

Keep the meaning precise: Action making product acceptable for intended use without necessarily meeting every original requirement.

B Precision and professional shorthand

EXTEND YOUR RANGE

reinspection

Inspection after a corrective operation or under a defined repeat process.

complete reinspection

material reconciliation

Accounting for all material quantities across defined states.

perform material reconciliation

mutually exclusive categories

Groups defined so that an item belongs to only one of them.

use mutually exclusive categories

double-counting

Including the same item more than once in a total.

prevent double-counting

defect count

The number of individual defects, potentially more than one per unit.

distinguish defect count

unit count

The number of separate items in the measured population.

verify the unit count

cost of poor quality

Costs arising because products or processes fail to meet requirements.

assess cost of poor quality

scrap cost

The defined cost assigned to scrapped product or material.

calculate scrap cost

rework labor

Labor used to bring product into conformity through rework.

record rework labor

recovery assumption

A premise about the quantity expected to become acceptable.

qualify the recovery assumption

hold status

A controlled state preventing use pending the required decision.

maintain hold status

quality escape

Nonconforming product that passes beyond its intended detection point.

investigate a quality escape

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

acceptance status ACCEPTANCE STATUS

verify acceptance status

Keep the meaning precise: Whether the required acceptance decision has been completed.

defect count DEFECT COUNT

distinguish defect count

Keep the meaning precise: The number of individual defects, potentially more than one per unit.

C Phrases that move the work forward

Define the population
The report covers 1,000 inspected units from this batch.

State the first-pass result
Nine hundred met requirements on the first inspection.

Separate the pending quantity
Sixty units await authorized rework.

State the scrap decision
Forty units have approved scrap disposition.

Reconcile the categories
These three exclusive states account for the full batch.

Correct the scrap total
The 100 non-first-pass units are not all scrap.

Keep recovery conditional
If all 60 pass after rework, the accepted total would be 960.

Report the current state
No reworked units have been accepted yet.

Name the yield basis
First-pass yield is 90% of the inspected population.

Name the scrap basis
The approved scrap quantity is 4% of these 1,000 units.

Distinguish defects and units
Two defects on one item are not two defective units.

Avoid double-counting
Move accepted reworked units between states rather than add them twice.

Separate costs
Scrap cost and rework labor are different cost categories.

Keep authorization visible
A rework plan does not remove the need for the required acceptance decision.

Correct the forecast label
Mark 960 as conditional recovery, not actual good output.

Close the report
Show current accepted, pending, and scrap quantities separately.

Precision in real exchanges

Purpose: State the scrap decision
Useful wording: Forty units have approved scrap disposition.

Purpose: Reconcile the categories
Useful wording: These three exclusive states account for the full batch.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which report reconciles the batch correctly?

A. 960 accepted now and 100 scrap

B. 900 accepted and 100 confirmed scrap

C. 1,000 accepted because rework is planned

D. 900 first-pass accepted, 60 awaiting rework, and 40 scrap

2. If all 60 later pass the required acceptance process, what would the accepted total be?

A. 960, making the original first-pass yield 96%

B. 1,060 because the rework count is added twice

C. 900 because reworked units can never be accepted

D. 960, while first-pass yield remains 90%

3. A colleague needs the term for this meaning: Product designated not to be used for its original intended purpose. Which term fits?

A. rework queue

B. scrap

C. rework labor

D. mutually exclusive categories

D Choose the response that fits / continued

4. Rebuild the exchange from "One thousand units, three different states". Ellis: "The dashboard also shows ninety-six percent first-pass yield. Someone added the sixty units they expect to recover, even though none has passed after rework yet." Priya: [missing response] Ellis: "Then the current accepted count is nine hundred. I can show a possible nine hundred sixty separately, but not as output already available." Which response belongs in the gap?

A. Wait for the required reinspection and acceptance decision. A finished operation does not by itself establish that the units meet requirements.

B. Retain rework here: restoring conformity to the requirements. Repair can have a different acceptance basis, so the terms are not interchangeable.

C. Correct. Sixty remain in the rework queue. Completion of the planned work and the required acceptance are still ahead of them.

D. The first-pass yield is ninety percent. Nine hundred passed at first inspection. A later recovery does not change that original first-pass result.

Answers and explanations, page 118

> **One useful distinction**
>
> 900 first-pass units plus 60 awaiting rework plus 40 scrap equals 1,000 inspected units. First-pass yield is 90%; approved scrap is 4% of this population. The remaining 6% is pending rework, not already recovered output.

E Conversation

One thousand units, three different states

One thousand units, three different states

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

scrap / rework / material reconciliation / rework queue / mutually exclusive categories / double-counting / defect count / recovery assumption / reinspection / first-pass yield

01 **Ellis:** The headline says one hundred units scrapped. The detail says nine hundred passed first time, sixty await rework, and forty are scrap. Which number should I report?

02 **Priya:** Forty is the approved [1] _____ quantity. The other sixty have not been discarded; they are awaiting a different process with its outcome still open.

03 **Ellis:** Let me check the batch total: nine hundred plus sixty plus forty is one thousand. Those categories cover every inspected unit once.

04 **Priya:** Yes, they are [2] _____. Keep the pending units out of both current accepted output and the scrap total.

05 **Ellis:** The dashboard also shows ninety-six percent first-pass yield. Someone added the sixty units they expect to recover, even though none has passed after rework yet.

06 **Priya:** The [3] _____ is ninety percent. Nine hundred passed at first inspection. A later recovery does not change that original first-pass result.

07 **Ellis:** Then the current accepted count is nine hundred. I can show a possible nine hundred sixty separately, but not as output already available.

08 **Priya:** Correct. Sixty remain in the [4] _____. Completion of the planned work and the required acceptance are still ahead of them.

09 **Ellis:** Would this forecast work: if all sixty pass after the authorized process, final accepted output would be nine hundred sixty?

10 **Priya:** That is a clear [5] _____. Keep the if and would; otherwise the forecast reads like a completed recovery.

Conversation continues on the next page.

One thousand units, three different states

CONVERSATION / CONTINUED

11 **Ellis:** The technician called the work repair in an email. Should I use that word in the report, or retain the process named in the record?

12 **Priya:** Retain [6] _____ here: restoring conformity to the requirements. Repair can have a different acceptance basis, so the terms are not interchangeable.

13 **Ellis:** When the work finishes, should I move the units straight into accepted output? The work-order status will tell me that the operation is complete.

14 **Priya:** Wait for the required [7] _____ and acceptance decision. A finished operation does not by itself establish that the units meet requirements.

15 **Ellis:** Another table lists three defects on one unit. I need to keep that from becoming three defective units in the batch reconciliation.

16 **Priya:** Exactly. The [8] _____ can be higher than the defective-unit count. One physical unit can contribute several separate defects.

17 **Ellis:** And when a unit is accepted after rework, it leaves the pending category. Otherwise our total would grow without any additional material.

18 **Priya:** That would be [9] _____. Move the status when the acceptance is recorded; do not leave the same unit in both balances.

19 **Ellis:** I will show nine hundred currently accepted, sixty pending, and forty scrap. Finance will need the cost basis before turning those quantities into a monetary loss.

20 **Priya:** Complete that [10] _____ first. The states are known; equal cost per scrap and rework unit is not one of our supplied facts.

CHECK THE GAPS

1. scrap / **2.** mutually exclusive categories / **3.** first-pass yield / **4.** rework queue / **5.** recovery assumption / **6.** rework / **7.** reinspection / **8.** defect count / **9.** double-counting / **10.** material reconciliation

Why these answers fit, page 119

F Check accuracy and scope

One thousand units, three different states

Return to the case

Quality analyst Priya and production manager Ellis review 1,000 inspected units from one batch. Exactly 900 passed on the first inspection, 60 are awaiting authorized rework, and 40 have an approved scrap disposition. These categories are mutually exclusive and account for all 1,000 units. No reworked units have yet been accepted. A summary labels all 100 non-first-pass units as scrap and forecasts 960 good units as though they already exist. Priya must correct both the loss and recovery statements.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. One unit has three defects and is included in the sixty awaiting rework. How should the unit reconciliation treat it?

A. Three accepted units once the three corrective operations are planned

B. One pending unit, with three defects recorded in the separate defect count

C. Three pending units because each defect needs its own correction

D. One pending unit plus one scrap unit until rework is completed

2. Which financial statement is supported without further data?

A. Total poor-quality cost is exactly 100 dollars.

B. Planned rework has no labor cost.

C. Every non-first-pass unit has the same financial consequence.

D. Scrap quantity and pending rework are known; their complete costs are not supplied.

Answers and explanations, page 121

Listen for the distinction

Swap roles. Keep ninety-percent first-pass yield unchanged by possible later recovery; 960 accepted remains conditional.

Make the categories reconcile

A clear quantity discussion starts with the population and the mutually exclusive states. Separate completed results from expected recovery. This prevents a commercially attractive forecast from entering the good-unit count and avoids exaggerating waste by labeling all unfinished work as irreversible loss.

F Say it / transfer to a new case

Keep future recovery conditional

A batch of 200 has 170 first-pass accepted units, 20 awaiting rework, and 10 approved scrap. No rework outcome is available.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

conditional / ten / eighty-five / twenty

01 **Analyst:** First-pass yield is [1] _____ percent.

02 **Manager:** The pending rework quantity is [2] _____.

03 **Analyst:** Approved scrap is [3] _____ units.

04 **Manager:** A total of 190 accepted units remains [4] _____.

CHECK THE GAPS

1. eighty-five / **2.** twenty / **3.** ten / **4.** conditional

Why these answers fit, page 121

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "One thousand units, three different states". Ellis: "When the work finishes, should I move the units straight into accepted output? The work-order status will tell me that the operation is complete." Priya: [missing response] Ellis: "Another table lists three defects on one unit. I need to keep that from becoming three defective units in the batch reconciliation." Which response belongs in the gap?

A. That would be double-counting. Move the status when the acceptance is recorded; do not leave the same unit in both balances.

B. Wait for the required reinspection and acceptance decision. A finished operation does not by itself establish that the units meet requirements.

C. Correct. Sixty remain in the rework queue. Completion of the planned work and the required acceptance are still ahead of them.

D. Retain rework here: restoring conformity to the requirements. Repair can have a different acceptance basis, so the terms are not interchangeable.

Answers and explanations, page 122

A Read the situation

LESSON 04 / ROOT CAUSE AND CORRECTIVE ACTION

Challenge a premature root-cause statement, preserve competing hypotheses, and distinguish containment from verified corrective action.

> **The supplier changed, but so did the process**
>
> Process engineer Jun and quality lead Rosa investigate a fictional surface defect. It first appears in the records after a supplier switch, but two process settings also changed during the same period. Comparable lot and setting records have not yet been reconciled. A draft names the supplier as the confirmed root cause and calls temporary sorting a completed corrective action. Jun and Rosa must organize the evidence, keep the cause open, and distinguish immediate screening from work that prevents recurrence.

People: Jun, Process engineer; Rosa, Quality lead.

Read the record

Choose one answer per question using the facts above.

1. What changed in the relevant period?

A. No identified process input
B. The supplier and two process settings
C. Only the supplier
D. Only the inspection report's color

2. What evidence is still unreconciled?

A. Comparable lot and process-setting records
B. A completed proof that every setting was constant
C. A final supplier admission
D. An approved effectiveness result

3. What is the status of temporary sorting?

A. A demonstrated permanent cause elimination
B. Evidence that the supplier definitely caused the defect
C. A completed effectiveness check for every future lot
D. An immediate screening measure, not proof of recurrence prevention

Answers and explanations, page 123

B Vocabulary in use

CORE TERMS

root cause
An underlying cause whose removal addresses the problem's recurrence.

establish the root cause

symptom
An observable sign of a problem rather than its underlying explanation.

distinguish a symptom from a cause

hypothesis
A proposed explanation that can be examined against evidence.

test a hypothesis

correlation
An observed association that does not by itself establish causation.

distinguish correlation from causation

confounding factor
Another changing variable that can affect an apparent relationship.

identify confounding factors

change history
A record of relevant modifications over time.

reconstruct the change history

process parameter
A controllable or measured variable describing process operation.

review process parameters

lot traceability
The ability to link material and records to particular lots.

maintain lot traceability

comparative evidence
Information allowing relevant conditions or groups to be compared.

obtain comparative evidence

stratification
Separating data into meaningful groups for analysis.

stratify by supplier and setting

cause-and-effect diagram
A structured display of possible contributors to a problem.

build a cause-and-effect diagram

five whys
A questioning method used to explore successive levels of explanation.

use five whys carefully

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

root cause ROOT CAUSE

establish the root cause

Keep the meaning precise: An underlying cause whose removal addresses the problem's recurrence.

change history CHANGE HISTORY

reconstruct the change history

Keep the meaning precise: A record of relevant modifications over time.

B Precision and professional shorthand

EXTEND YOUR RANGE

8D
An eight-discipline structured approach to team problem solving.
prepare an 8D report

sorting
Separating items using defined inspection or classification criteria.
document temporary sorting

occurrence cause
A cause explaining why the defect was produced.
verify the occurrence cause

corrective action
Action addressing a cause to prevent recurrence.
implement corrective action

recurrence
The same or a related problem appearing again.
monitor recurrence

controlled comparison
A planned comparison that accounts for relevant differing conditions.
authorize a controlled comparison

containment action
An immediate measure intended to limit the impact of a problem.
implement a containment action

escape point
The point at which a problem should have been detected but was not.
identify the escape point

detection failure
A reason an existing check did not identify the problem.
investigate a detection failure

effectiveness check
Evaluation of whether an action achieved its intended result.
define an effectiveness check

verification evidence
Recorded support that a specified condition or action has been achieved.
retain verification evidence

closure criteria
Defined conditions for accepting an issue as resolved.
agree closure criteria

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

five whys FIVE WHYS

use five whys carefully

Keep the meaning precise: A questioning method used to explore successive levels of explanation.

occurrence cause OCCURRENCE CAUSE

verify the occurrence cause

Keep the meaning precise: A cause explaining why the defect was produced.

C Phrases that move the work forward

State the timing

The defect first appears after the supplier switch in the available records.

Qualify the explanation

The supplier is one hypothesis, not a confirmed root cause.

Separate association

The timing suggests a question; it does not establish causation.

Distinguish prevention

Sorting does not by itself prevent the defect from being produced.

Avoid an unsupported accusation

Ask for supplier records without stating a cause we have not established.

Bound further work

Any comparison or trial must follow the authorized process.

Define effectiveness

Specify what evidence would show the action prevents recurrence.

Assign the next review

Jun will reconcile the change and lot records for the next review.

Identify competing changes

Two process settings changed during the same period.

Request comparable records

Link the defect data to the actual supplier lots and settings.

Describe containment

Temporary sorting screens output while the investigation continues.

Keep the evidence organized

Stratify the observations by the relevant conditions.

Check detection separately

Why did the existing check fail to identify the issue earlier?

Do not promote a diagram

A listed possible cause is not a verified finding.

Keep closure open

The issue remains open until the agreed criteria are met.

Close with status

Containment is active; cause and corrective-action effectiveness remain unconfirmed.

Precision in real exchanges

Purpose: Request comparable records
Useful wording: Link the defect data to the actual supplier lots and settings.

Purpose: Separate association
Useful wording: The timing suggests a question; it does not establish causation.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which root-cause statement fits the evidence?

A. The supplier is proven responsible because its change came first.

B. The settings are proven responsible because there are two of them.

C. No investigation is possible when several things change.

D. The supplier change and two setting changes remain competing explanations pending comparison.

2. What does sorting establish by itself?

A. The occurrence cause is eliminated.

B. Every future lot will be defect-free.

C. The supplier admitted responsibility.

D. A screening activity is being used; recurrence prevention is not demonstrated.

3. A colleague needs the term for this meaning: An observed association that does not by itself establish causation. Which term fits?

A. sorting

B. comparative evidence

C. recurrence

D. correlation

D Choose the response that fits / continued

4. Rebuild the exchange from "The supplier changed, but so did the process". Rosa: "The process settings could affect the result directly or interact with the supplied material. The current summary treats them as background details instead of relevant changes." Jun: [missing response] Rosa: "What should the evidence package contain? The identifiers in our defect, receiving, and machine records do not yet line up." Which response belongs in the gap?

A. That is where lot traceability matters. Match the actual lots, settings, and observations before assuming that every unit made after the announcement used the new material.

B. Yes, examine the detection failure on its own evidence. Do not assume that identifying a missed check also identifies the mechanism that created the defect.

C. It does not yet. The supplier is a hypothesis. Two process settings changed in the same period, and the comparable lot records have not been reconciled.

D. Each is a possible confounding factor. If conditions changed together, we cannot attribute the outcome to the supplier alone without further supporting comparison.

Answers and explanations, page 123

One useful distinction

After the supplier switch describes timing, not a proven causal relationship. Two settings also changed, so those variables can confound the apparent supplier effect. Reconcile comparable records before treating one explanation as established.

E Conversation

The supplier changed, but so did the process

The supplier changed, but so did the process

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

change history / detection failure / occurrence cause / confounding factor / closure criteria / containment action / correlation / effectiveness check / lot traceability / hypothesis

01 **Rosa:** The draft says the new supplier is the root cause because the defect first appeared after the switch. I want to check whether the evidence really isolates that change.

02 **Jun:** It does not yet. The supplier is a [1] _____. Two process settings changed in the same period, and the comparable lot records have not been reconciled.

03 **Rosa:** I will keep the dates and remove the confirmed-cause label. Which competing explanations do you want the investigation team to check?

04 **Jun:** Exactly. The observed [2] _____ does not establish which change produced the defect. We need to examine the competing explanations rather than select the easiest one to describe.

05 **Rosa:** The process settings could affect the result directly or interact with the supplied material. The current summary treats them as background details instead of relevant changes.

06 **Jun:** Each is a possible [3] _____. If conditions changed together, we cannot attribute the outcome to the supplier alone without further supporting comparison.

E Conversation / continued

07 **Rosa:** What should the evidence package contain? The identifiers in our defect, receiving, and machine records do not yet line up.

08 **Jun:** Reconstruct the [4] _____ and link it to actual production. Dates in a meeting summary are not enough if the material or settings took effect at different times.

09 **Rosa:** Can you link each run to the actual material lot? The monthly supplier summary will not tell us whether the old stock was still being used.

10 **Jun:** That is where [5] _____ matters. Match the actual lots, settings, and observations before assuming that every unit made after the announcement used the new material.

Conversation continues on the next page.

The supplier changed, but so did the process

CONVERSATION / CONTINUED

11 **Rosa:** The sorting team has been screening output while we investigate. The draft calls that corrective action complete, which sounds stronger than the work establishes.

12 **Jun:** Call it a [6] _____. Sorting may limit the movement of affected output, but it does not by itself stop the process from producing the defect.

13 **Rosa:** We should still evaluate the sorting method through the appropriate procedure. A temporary measure should not be treated as perfect simply because it is currently in place.

14 **Jun:** Agreed. And keep the [7] _____ separate from the screening question. One concerns why the defect is created; the other concerns whether the existing controls find it.

15 **Rosa:** That leaves a second investigation line: why the previous check did not flag the issue. Solving that could improve detection even before the production explanation is settled.

16 **Jun:** Yes, examine the [8] _____ on its own evidence. Do not assume that identifying a missed check also identifies the mechanism that created the defect.

17 **Rosa:** Once the evidence supports an action, we need a defined way to judge whether it worked. A completed supplier response form will not be enough.

18 **Jun:** Set an [9] _____ with the appropriate criteria and review period. Any trial or process change must follow the authorized procedure rather than become an improvised production experiment.

19 **Rosa:** I will revise the status: containment active, supplier and setting hypotheses open, and record reconciliation assigned to you. We will not announce a cause before the evidence supports it.

20 **Jun:** Good. Keep the [10] _____ visible so the team knows what remains. A clear open status is more useful than a closed label built on an untested explanation.

CHECK THE GAPS

1. hypothesis / **2.** correlation / **3.** confounding factor / **4.** change history / **5.** lot traceability / **6.** containment action / **7.** occurrence cause / **8.** detection failure / **9.** effectiveness check / **10.** closure criteria

Why these answers fit, page 124

F Check accuracy and scope

The supplier changed, but so did the process

Return to the case

Process engineer Jun and quality lead Rosa investigate a fictional surface defect. It first appears in the records after a supplier switch, but two process settings also changed during the same period. Comparable lot and setting records have not yet been reconciled. A draft names the supplier as the confirmed root cause and calls temporary sorting a completed corrective action. Jun and Rosa must organize the evidence, keep the cause open, and distinguish immediate screening from work that prevents recurrence.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which request best tests the competing supplier and setting explanations?

A. Group every later defect under the new supplier without checking setting history.

B. Count all proposed causes and select the one mentioned most often.

C. Treat completion of sorting as verification that the supplier caused the defect.

D. Link defect observations to actual supplier lots and both settings under comparable conditions.

2. What distinguishes an occurrence cause from a detection failure?

A. One explains production of the defect; the other explains why a check missed it.

B. Both always mean the supplier changed.

C. A missed check proves why the defect was created.

D. Eliminating a detection failure automatically eliminates occurrence.

Answers and explanations, page 126

Listen for the distinction

Swap roles. Distinguish sorting from recurrence prevention and the occurrence cause from a missed detection.

A hypothesis is not an accusation

Use conditional language to keep plausible explanations available for testing. This is not a refusal to decide; it identifies what evidence a decision requires. A supplier discussion can be direct about the observed defect without presenting timing alone as proof of responsibility.

F Say it / transfer to a new case

Two changes, no isolated cause

A defect appears after a material change and a temperature-setting change. No comparative analysis is complete. Temporary sorting is active.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

action / factor / evidence / hypothesis

01 **Engineer:** The material explanation is still a [1] _____.

02 **Reviewer:** The setting change is a potential confounding [2] _____.

03 **Engineer:** Temporary sorting is a containment [3] _____.

04 **Reviewer:** Recurrence prevention still needs effectiveness [4] _____.

CHECK THE GAPS

1. hypothesis / **2.** factor / **3.** action / **4.** evidence

Why these answers fit, page 126

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "The supplier changed, but so did the process". Rosa: "We should still evaluate the sorting method through the appropriate procedure. A temporary measure should not be treated as perfect simply because it is currently in place." Jun: [missing response] Rosa: "That leaves a second investigation line: why the previous check did not flag the issue. Solving that could improve detection even before the production explanation is settled." Which response belongs in the gap?

A. Exactly. The observed correlation does not establish which change produced the defect. We need to examine the competing explanations rather than select the easiest one to describe.

B. Reconstruct the change history and link it to actual production. Dates in a meeting summary are not enough if the material or settings took effect at different times.

C. It does not yet. The supplier is a hypothesis. Two process settings changed in the same period, and the comparable lot records have not been reconciled.

D. Agreed. And keep the occurrence cause separate from the screening question. One concerns why the defect is created; the other concerns whether the existing controls find it.

Answers and explanations, page 127

Review 2 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Defects, Scrap, and Rework" and "Root Cause and Corrective Action". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: A unit that does not meet a specified requirement. Which term fits?

A. final yield
B. material reconciliation
C. scrap cost
D. nonconforming unit

2. A colleague needs the term for this meaning: The ability to link material and records to particular lots. Which term fits?

A. root cause
B. lot traceability
C. 8D
D. corrective action

3. Rebuild the exchange from "The queue is for a shared tool". Marcus: "That would also help us separate people walking to retrieve the tool from material being moved between work areas. Both happen, but they are different kinds of movement." Ana: [missing response] Marcus: "For the trial, the suggestion is to keep an appropriate tool at this one bench. We have not decided that every work area needs a new tool arrangement." Which response belongs in the gap?

A. That is a proposal for point-of-use storage. Check the practical requirements and shared demand before assuming that moving the tool solves the problem for the whole process.
B. Make that explicit in the pilot trial description. Use comparable task conditions and the same measure, and record any new difficulties rather than reporting only the favorable observations.
C. Yes. Unnecessary motion concerns the people performing the task. We should describe the actual movement rather than use a broad waste label that hides what needs to change.
D. Then name the observed waiting. The tool was unavailable at those moments; that does not establish that the operators worked slowly when they had what they needed.

Review 2 / Meaning and context / continued

4. Rebuild the exchange from "The supplier changed, but so did the process". Rosa: "Once the evidence supports an action, we need a defined way to judge whether it worked. A completed supplier response form will not be enough." Jun: [missing response] Rosa: "I will revise the status: containment active, supplier and setting hypotheses open, and record reconciliation assigned to you. We will not announce a cause before the evidence supports it." Which response belongs in the gap?

A. Good. Keep the closure criteria visible so the team knows what remains. A clear open status is more useful than a closed label built on an untested explanation.

B. Set an effectiveness check with the appropriate criteria and review period. Any trial or process change must follow the authorized procedure rather than become an improvised production experiment.

C. Exactly. The observed correlation does not establish which change produced the defect. We need to examine the competing explanations rather than select the easiest one to describe.

D. Reconstruct the change history and link it to actual production. Dates in a meeting summary are not enough if the material or settings took effect at different times.

Answers and explanations, page 151

A Read the situation

LESSON 05 / MAINTENANCE, RELIABILITY, AND CHANGEOVER

Explain schedule conflicts, distinguish work types, and present conditional options without assuming authority to defer maintenance or restart equipment.

A rush order meets the maintenance window

Maintenance planner Leila and production scheduler Ben face a conflict. A planned two-hour maintenance window runs from 13:00 to 15:00, while a new rush order would need the same line through 14:00 under the current schedule. No technical assessment or authorization to defer maintenance is supplied. An alternative line might be available, but its suitability is unconfirmed. Leila and Ben must prepare options for the responsible decision maker without describing a possible workaround as an approved plan.

People: Leila, Maintenance planner; Ben, Production scheduler.

Read the record

Choose one answer per question using the facts above.

1. When do the two demands overlap?

A. They do not overlap
B. From 13:00 to 14:00
C. From 14:00 to 15:00 only
D. For the entire day

2. What is known about maintenance deferral?

A. It is automatically safe because the order is urgent.
B. Production has already authorized it.
C. Maintenance has no effect on reliability or safety.
D. No supporting technical assessment or authorization is supplied.

3. What is the alternative-line status?

A. Fully qualified and reserved
B. Proven to need no changeover
C. Already producing the order
D. Possible but unconfirmed in suitability

Answers and explanations, page 128

B Vocabulary in use

CORE TERMS

preventive maintenance
Planned work intended to reduce the likelihood of equipment failure.

schedule preventive maintenance

predictive maintenance
Maintenance informed by indicators of equipment condition or future failure.

use predictive maintenance

corrective maintenance
Work performed to restore function after a fault is identified.

plan corrective maintenance

condition monitoring
Tracking indicators of equipment health over time.

review condition-monitoring data

maintenance window
A reserved period for planned maintenance activity.

protect the maintenance window

work order
A controlled record authorizing and describing a defined maintenance task.

review the work order

backlog
Approved or identified work that remains unfinished.

prioritize the maintenance backlog

critical asset
Equipment whose failure has significant operational or other consequences.

identify critical assets

reliability
The ability to perform the required function over a stated period under stated conditions.

assess equipment reliability

maintainability
The ease and speed with which equipment can be maintained or restored.

improve maintainability

mean time between failures
Average operating time between failures for the defined repairable population.

interpret mean time between failures

mean time to repair
Average repair time under the stated measurement definition.

state the mean-time-to-repair basis

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

preventive maintenance PREVENTIVE MAINTENANCE

schedule preventive maintenance

Keep the meaning precise: Planned work intended to reduce the likelihood of equipment failure.

work order WORK ORDER

review the work order

Keep the meaning precise: A controlled record authorizing and describing a defined maintenance task.

B Precision and professional shorthand

EXTEND YOUR RANGE

failure mode

The way in which an item fails to perform its function.

identify a failure mode

wear indicator

A measured or observed sign of component deterioration.

monitor wear indicators

spare-part readiness

Availability of the correct parts needed for the planned work.

confirm spare-part readiness

changeover

Transitioning equipment from one product or setup to another.

plan the changeover

setup time

The time required to prepare equipment for a specified job.

measure setup time

SMED

Single-minute exchange of die: a method for reducing changeover time.

apply SMED principles

internal setup

Setup activity that requires the machine to be stopped.

identify internal setup tasks

external setup

Setup activity that can be completed while the machine is running.

separate external setup tasks

deferral

Postponement of a planned task subject to the required assessment and decision.

assess a maintenance deferral

return to service

The authorized restoration of equipment to operational use.

confirm return to service

capacity contingency

A conditional alternative for meeting production needs.

assess a capacity contingency

schedule dependency

A condition or predecessor affecting the timing of planned work.

identify schedule dependencies

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

mean time to repair MEAN TIME TO REPAIR

state the mean-time-to-repair basis

Keep the meaning precise: Average repair time under the stated measurement definition.

setup time SETUP TIME

measure setup time

Keep the meaning precise: The time required to prepare equipment for a specified job.

C Phrases that move the work forward

State the overlap

The order needs the line until 14:00, while maintenance starts at 13:00.

Name the open assessment

We do not yet have a basis for deferring the maintenance.

Present a conditional option

An alternative line may be possible if suitability and readiness are confirmed.

Separate work types

This is maintenance, not merely a product changeover.

Request readiness information

Confirm the work scope, required parts, personnel, and dependencies.

Qualify the finish time

The plan needs its conditions stated before we promise completion.

Offer commercial alternatives

The decision maker can also assess a revised delivery commitment.

Communicate the effect

Tell affected teams which line and time assumptions changed.

Keep duration distinct

A one-hour conflict does not mean a two-hour task can be completed in one hour.

Respect decision ownership

The responsible decision maker needs the technical assessment before approving a change.

Check more than space

An idle line is not automatically qualified for this order.

Preserve required scope

Do not remove required work just to fit the schedule.

Avoid a reliability guarantee

Recent operation does not prove the task can be postponed without consequence.

Keep return distinct

Finishing the work is not the same as authorized return to service.

Record the actual decision

Update the schedule only after the authorized choice is documented.

Close with next steps

Present the options and unresolved conditions together.

Precision in real exchanges

Purpose: Respect decision ownership
Useful wording: The responsible decision maker needs the technical assessment before approving a change.

Purpose: Present a conditional option
Useful wording: An alternative line may be possible if suitability and readiness are confirmed.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which schedule statement is accurate?

A. There is a one-hour overlap, but the maintenance scope remains two hours unless properly reassessed.

B. The maintenance must take only one hour because the overlap is one hour.

C. The order and maintenance never conflict.

D. The rush order automatically cancels the maintenance.

2. Which alternative-line statement is best?

A. A possible alternative is already an approved schedule.

B. Assess suitability, setup, readiness, and allocation before committing the order.

C. Any idle line can run any product.

D. Availability alone proves qualification.

3. A colleague needs the term for this meaning: Tracking indicators of equipment health over time. Which term fits?

A. spare-part readiness

B. condition monitoring

C. deferral

D. reliability

D Choose the response that fits / continued

4. Rebuild the exchange from "A rush order meets the maintenance window". Ben: "The customer will ask why we cannot use a machine that ran yesterday. What can I say without suggesting that yesterday's run settles today's maintenance decision?" Leila: [missing response] Ben: "What information should we bring to the decision maker? I can show the order timing and the impact on delivery, but the work scope needs your input." Which response belongs in the gap?

A. Exactly. External setup refers to eligible tasks that can occur while equipment runs. Required safety, quality, and operating controls still govern how the work is organized.

B. We should review the work order, required resources, and relevant dependencies. The schedule discussion must not silently delete part of the task to make the overlap disappear.

C. Treat deferral as an option for assessment, not an approved decision. We need the responsible review before changing the commitment or telling the team to proceed.

D. It does not establish reliability for the proposed period. The task's purpose, equipment condition, and consequences need assessment rather than a conclusion based only on yesterday's operation.

Answers and explanations, page 128

One useful distinction

The conflict is one hour, from 13:00 to 14:00, but moving a two-hour maintenance task may have consequences beyond that overlap. Its duration does not establish that it can be shortened, split, or deferred without the required assessment.

E Conversation

A rush order meets the maintenance window

A rush order meets the maintenance window

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

maintenance window / capacity contingency / schedule dependency / reliability / work order / deferral / changeover / External setup / Return to service / spare-part readiness

01 **Ben:** The rush order needs the line until fourteen hundred. Maintenance is booked from thirteen hundred to fifteen hundred, so the current schedule cannot satisfy both plans.

02 **Leila:** The [1] _____ overlaps the order by one hour. That does not mean the two-hour task can be shortened to one hour.

03 **Ben:** Could we move the work later? I need an option for the customer discussion, but I do not have the technical basis to say that postponement is acceptable.

04 **Leila:** Treat [2] _____ as an option for assessment, not an approved decision. We need the responsible review before changing the commitment or telling the team to proceed.

05 **Ben:** The customer will ask why we cannot use a machine that ran yesterday. What can I say without suggesting that yesterday's run settles today's maintenance decision?

06 **Leila:** It does not establish [3] _____ for the proposed period. The task's purpose, equipment condition, and consequences need assessment rather than a conclusion based only on yesterday's operation.

E Conversation / continued

07 **Ben:** What information should we bring to the decision maker? I can show the order timing and the impact on delivery, but the work scope needs your input.

08 **Leila:** We should review the [4] _____, required resources, and relevant dependencies. The schedule discussion must not silently delete part of the task to make the overlap disappear.

09 **Ben:** Line two may be free this afternoon. I will check its product capability and setup requirements before offering it as the alternative.

10 **Leila:** Call it a [5] _____. An unoccupied line is not automatically suitable, ready, or allocated to this order, so we cannot promise that route yet.

Conversation continues on the next page.

A rush order meets the maintenance window

CONVERSATION / CONTINUED

11 **Ben:** It would need a different product setup. That is separate from the maintenance on the original line, even though both activities take equipment time.

12 **Leila:** Correct. A [6] _____ prepares the alternative line for the job. Its scope and timing need review; it does not replace the original maintenance requirement.

13 **Ben:** The improvement team has discussed moving preparation outside stopped-machine time. We should not assume every preparation activity can be moved that way.

14 **Leila:** Exactly. [7] _____ refers to eligible tasks that can occur while equipment runs. Required safety, quality, and operating controls still govern how the work is organized.

15 **Ben:** We also need the correct maintenance parts and people available. A calendar reservation alone does not demonstrate that the work can finish as planned.

16 **Leila:** Confirm [8] _____ and the other resources. Missing items could change the plan, and the decision maker needs those facts alongside the customer timing.

17 **Ben:** Our options are therefore conditional: assess the alternative line, assess a permitted schedule change, or discuss a revised delivery commitment. None is confirmed yet.

18 **Leila:** Yes. Show each [9] _____ and the responsible owner. That makes the trade-off visible without allowing the rush order to become an unsupported technical authorization.

19 **Ben:** Once the authorized choice is recorded, I will update the affected teams. I will also avoid describing the equipment as ready merely because the work is reported finished.

20 **Leila:** Good. [10] _____ remains a distinct status. The final update should identify the completed work, required checks, and actual authorization before making a production commitment.

CHECK THE GAPS

1. maintenance window / **2.** deferral / **3.** reliability / **4.** work order / **5.** capacity contingency / **6.** changeover / **7.** External setup / **8.** spare-part readiness / **9.** schedule dependency / **10.** Return to service

Why these answers fit, page 130

F Check accuracy and scope

A rush order meets the maintenance window

Return to the case

Maintenance planner Leila and production scheduler Ben face a conflict. A planned two-hour maintenance window runs from 13:00 to 15:00, while a new rush order would need the same line through 14:00 under the current schedule. No technical assessment or authorization to defer maintenance is supplied. An alternative line might be available, but its suitability is unconfirmed. Leila and Ben must prepare options for the responsible decision maker without describing a possible workaround as an approved plan.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which reply preserves authority?

A. A planner's suggestion is sufficient return-to-service approval.
B. We can present deferral for assessment, but it is not authorized in the current information.
C. We can skip it because production has a deadline.
D. No assessment is needed if the machine is currently running.

2. The maintenance operation is marked finished. What remains a separate status to confirm?

A. Whether the original window lasted two hours
B. Whether the order and maintenance once overlapped
C. Whether the planner had mentioned an alternative line
D. Required checks and authorized return to service

Answers and explanations, page 131

Listen for the distinction

Swap roles. Present the other line as conditional, and distinguish proposed deferral from authorization.

Acknowledge urgency without borrowing authority

A useful maintenance discussion separates customer pressure from the technical assessment and the authorized decision. State the conflict precisely and present options with their conditions. Avoid a casual we can just move it when the people in the conversation have not established that deferral is acceptable.

F Say it / transfer to a new case

An option still has conditions

Maintenance is planned from 10:00 to 12:00. An order needs the same equipment until 11:00. A second line is possible but its suitability is unconfirmed.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

one / contingency / deferral / suitability

01 **Planner:** The schedules overlap by [1] _____ hour.

02 **Technician:** Moving maintenance would be a proposed [2] _____.

03 **Planner:** The second line is a capacity [3] _____.

04 **Technician:** Before using it, confirm its [4] _____.

CHECK THE GAPS

1. one / **2.** deferral / **3.** contingency / **4.** suitability

Why these answers fit, page 131

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "A rush order meets the maintenance window". Ben: "The improvement team has discussed moving preparation outside stopped-machine time. We should not assume every preparation activity can be moved that way." Leila: [missing response] Ben: "We also need the correct maintenance parts and people available. A calendar reservation alone does not demonstrate that the work can finish as planned." Which response belongs in the gap?

A. Good. Return to service remains a distinct status. The final update should identify the completed work, required checks, and actual authorization before making a production commitment.

B. Exactly. External setup refers to eligible tasks that can occur while equipment runs. Required safety, quality, and operating controls still govern how the work is organized.

C. It does not establish reliability for the proposed period. The task's purpose, equipment condition, and consequences need assessment rather than a conclusion based only on yesterday's operation.

D. Treat deferral as an option for assessment, not an approved decision. We need the responsible review before changing the commitment or telling the team to proceed.

Answers and explanations, page 132

A Read the situation

LESSON 06 / EHS AND SAFETY COMMUNICATION

Raise an ambiguous safety instruction clearly, distinguish shutdown from energy control, and confirm the authorized clarification route.

Stop at the unclear instruction

During a classroom safety exercise, operator Hana reads a training card saying switch off, then clean. Trainer Oscar confirms that the card does not identify the equipment, energy sources, or applicable isolation procedure. No actual equipment work is underway. Hana is not being authorized to service a machine. The exercise must stop at the ambiguous instruction and refer to the current approved equipment-specific procedure and qualified personnel. Hana and Oscar must clarify the communication without inventing an isolation sequence.

People: Hana, Operator in a classroom exercise; Oscar, Safety trainer.

Read the record

Choose one answer per question using the facts above.

1. Where is this exchange taking place?

A. In a classroom safety exercise with no equipment work underway
B. During an authorized live servicing operation
C. At a completed restart inspection
D. In a confirmed emergency rescue

2. Which clarification is needed before this classroom card could support the intended technical discussion?

A. A generic sequence assumed suitable for every machine
B. Identified equipment, energy sources, and the current applicable approved procedure
C. A statement that switching off is always equivalent to isolation
D. A learner signature substituting for the missing equipment reference

3. What must not be invented?

A. An isolation sequence for an unidentified machine
B. A clear statement that the card is ambiguous
C. A request for the approved procedure
D. A named clarification route

Answers and explanations, page 133

B Vocabulary in use

CORE TERMS

EHS
Environment, health, and safety functions and practices.

raise an EHS concern

hazard
A source or condition with the potential to cause harm.

identify a hazard

hazardous energy
Energy that can injure someone if unexpectedly released or applied.

recognize hazardous energy

energy isolation
Separation from energy sources using the applicable control method.

follow the energy-isolation procedure

lockout
Securing an energy-isolating device in the required safe position under the applicable procedure.

follow the lockout procedure

tagout
Use of a warning tag within an applicable energy-control system.

understand tagout limitations

stored energy
Energy remaining in a system after its normal supply is interrupted.

address stored energy

residual energy
Energy that remains after an initial control or shutdown action.

assess residual energy

shutdown
Stopping normal equipment operation.

distinguish shutdown from isolation

unexpected startup
Unintended equipment operation that can expose people to hazards.

prevent unexpected startup

authorized employee
A person assigned and trained for the relevant hazardous-energy-control work.

identify the authorized employee

affected employee
A person whose operation or work area is affected by energy-control activity.

inform affected employees

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

EHS EHS

raise an EHS concern

Keep the meaning precise: Environment, health, and safety functions and practices.

tagout TAGOUT

understand tagout limitations

Keep the meaning precise: Use of a warning tag within an applicable energy-control system.

B Precision and professional shorthand

EXTEND YOUR RANGE

equipment-specific procedure

Approved instructions applicable to identified equipment and work.

consult the equipment-specific procedure

stop-work concern

A safety concern requiring work to pause under the applicable process.

raise a stop-work concern

job hazard analysis

Review of job steps, associated hazards, and appropriate controls.

complete a job hazard analysis

personal protective equipment

Protective items worn to reduce exposure to specified hazards.

select required personal protective equipment

safety data sheet

A document describing a chemical's hazards and handling information.

consult the safety data sheet

read-back

Repeating critical information to check shared understanding.

request a read-back

verification of isolation

Confirmation of the required isolated state using the approved method.

require verification of isolation

near miss

An event that could have caused harm but did not in the observed outcome.

report a near miss

permit to work

A formal authorization process for defined work where required.

verify permit-to-work requirements

hierarchy of controls

An ordered approach to choosing hazard-control measures.

apply the hierarchy of controls

exclusion zone

An area with controlled access because of defined hazards or work.

respect the exclusion zone

restart authorization

The required permission to resume equipment operation.

confirm restart authorization

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

affected employee AFFECTED EMPLOYEE

inform affected employees

Keep the meaning precise: A person whose operation or work area is affected by energy-control activity.

job hazard analysis JOB HAZARD ANALYSIS

complete a job hazard analysis

Keep the meaning precise: Review of job steps, associated hazards, and appropriate controls.

C Phrases that move the work forward

State the concern

Stop here: the instruction does not identify the equipment or isolation procedure.

Distinguish the actions

Switching off is not the same as verifying energy isolation.

Request the controlled source

Use the current equipment-specific procedure.

Avoid improvisation

I will not infer the missing sequence from a generic instruction.

Ask for confirmation

Please confirm the document identity and current revision.

Avoid a false safe claim

We have no evidence here that any equipment is isolated.

Name the escalation

Oscar will refer the ambiguous card for correction through the training process.

Separate restart

A clarification does not itself authorize equipment restart.

Keep the pause clear

We should not continue from this ambiguous card.

Identify missing context

Which equipment and approved procedure does this instruction refer to?

Respect assigned roles

The relevant authorized personnel must handle the required assessment and work.

Keep training separate

This classroom exercise does not authorize actual servicing.

Use read-back

I will repeat the clarified scope and next step to check understanding.

Do not substitute protection

Protective equipment does not replace the required energy-control process.

Preserve the issue

Record the unclear wording so it is not reused unchanged.

Close the exercise

We will resume the classroom task only with the appropriate reviewed material.

Precision in real exchanges

Purpose: Identify missing context
Useful wording: Which equipment and approved procedure does this instruction refer to?

Purpose: Request the controlled source
Useful wording: Use the current equipment-specific procedure.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which response best fits the ambiguous card?

A. Ask an unassigned learner to certify the machine safe.

B. Pause and obtain the current applicable equipment-specific procedure through the authorized route.

C. Guess a generic isolation sequence from memory.

D. Continue because the words switch off guarantee isolation.

2. What does switching off establish by itself?

A. Restart is authorized.

B. Every possible energy source has been verified controlled.

C. Normal operation may be stopped; complete energy isolation is not established.

D. All stored energy is necessarily removed.

3. A colleague needs the term for this meaning: Separation from energy sources using the applicable control method. Which term fits?

A. safety data sheet

B. shutdown

C. stop-work concern

D. energy isolation

D Choose the response that fits / continued

4. Rebuild the exchange from "Stop at the unclear instruction". Hana: "There could also be energy remaining after a normal stop. I should not assume that the absence of movement proves there is nothing left that could cause harm." Oscar: [missing response] Hana: "Which machine is this card supposed to cover? Can we get its approved instructions before the class goes any further?" Which response belongs in the gap?

A. That is a clear stop-work concern. We are in a classroom exercise, and we should not continue as though the missing equipment information were already known.

B. Exactly. Stored energy needs the approved process for this equipment. We can't fill the gap by improvising an isolation procedure.

C. You are not. The authorized employee has an assigned role and the required training. Participating in this discussion does not give you that authorization.

D. Correct. Verification of isolation is a separate technical requirement under the applicable procedure. Our conversation does not establish that it has occurred.

Answers and explanations, page 133

> **One useful distinction**
>
> Stopping normal operation is not the same as controlling every hazardous energy source. This classroom card does not identify a machine or an approved method. The correct language response is to pause and seek the applicable procedure, not supply missing operational steps.

E Conversation

Stop at the unclear instruction

Stop at the unclear instruction

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

Stored energy / restart authorization / equipment-specific procedure / read-back / EHS / Verification of isolation / authorized employee / Shutdown / Tagout / stop-work concern

01 **Hana:** I want to stop at this line. The card says switch off, then clean, but it does not identify the machine or the procedure we should be using.

02 **Oscar:** That is a clear [1] _____. We are in a classroom exercise, and we should not continue as though the missing equipment information were already known.

03 **Hana:** The phrase switch off sounds simple, but it could refer only to stopping normal operation. It does not tell me what happens to other energy sources.

04 **Oscar:** Correct. [2] _____ is not the same as complete energy control. This card gives us no basis for declaring any machine isolated or safe for the proposed work.

05 **Hana:** There could also be energy remaining after a normal stop. I should not assume that the absence of movement proves there is nothing left that could cause harm.

06 **Oscar:** Exactly. [3] _____ needs the approved process for this equipment. We can't fill the gap by improvising an isolation procedure.

07 **Hana:** Which machine is this card supposed to cover? Can we get its approved instructions before the class goes any further?

08 **Oscar:** We need the current [4] _____ and the relevant qualified personnel. The correct document and assigned roles matter as much as the words on the card.

09 **Hana:** I understand the need to ask, but I do not want my question to sound as though I am volunteering to perform the isolation myself.

10 **Oscar:** You are not. The [5] _____ has an assigned role and the required training. Participating in this discussion does not give you that authorization.

Conversation continues on the next page.

Stop at the unclear instruction

CONVERSATION / CONTINUED

11 **Hana:** Some learners also treat a tag as though it physically prevents the same actions as a lock. We should make the distinction clear without demonstrating an unapproved method.

12 **Oscar:** Yes. [6] _____ has particular functions and limitations under the applicable system. A warning tag must not be described as providing identical physical restraint to a lock.

13 **Hana:** Once the correct information is available, I can repeat the equipment identity and the next communication step. That would help confirm that I heard the clarification correctly.

14 **Oscar:** That is [7] _____. It checks our shared understanding; it does not replace the required technical process or demonstrate that the equipment has reached a particular state.

15 **Hana:** The worksheet asks me to mark the machine isolated. I am leaving that blank: we are in a classroom and have not checked any equipment.

16 **Oscar:** Correct. [8] _____ is a separate technical requirement under the applicable procedure. Our conversation does not establish that it has occurred.

17 **Hana:** I will keep the concern specific: missing equipment identity and procedure. The training card needs correction, rather than a handwritten guess about the missing steps.

18 **Oscar:** I will refer it through the [9] _____ and training process. We should preserve the unclear wording so the responsible people can review and correct the source.

19 **Hana:** Then this step stays on hold until the reviewed instruction is available. I'll make sure nobody mistakes our clarification for permission to start cleaning.

20 **Oscar:** Exactly. It also provides no [10] _____. The outcome today is a clear concern, a proper referral, and an explicit boundary between classroom communication and actual equipment work.

CHECK THE GAPS

1. stop-work concern / **2.** Shutdown / **3.** Stored energy / **4.** equipment-specific procedure / **5.** authorized employee / **6.** Tagout / **7.** read-back / **8.** Verification of isolation / **9.** EHS / **10.** restart authorization

Why these answers fit, page 134

F Check accuracy and scope

Stop at the unclear instruction

Return to the case

During a classroom safety exercise, operator Hana reads a training card saying switch off, then clean. Trainer Oscar confirms that the card does not identify the equipment, energy sources, or applicable isolation procedure. No actual equipment work is underway. Hana is not being authorized to service a machine. The exercise must stop at the ambiguous instruction and refer to the current approved equipment-specific procedure and qualified personnel. Hana and Oscar must clarify the communication without inventing an isolation sequence.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. What can read-back confirm?

A. That no approved procedure is needed

B. Shared understanding of the clarified message, not the physical isolated state

C. That every machine is safe to clean

D. That the learner is automatically authorized

2. Which closing statement is accurate?

A. The equipment is now isolated because the discussion ended.

B. Every participant can now perform servicing independently.

C. The unclear card is acceptable if someone signs it.

D. The ambiguous classroom card is referred for correction; no equipment work or restart is authorized.

Answers and explanations, page 136

Listen for the distinction

Swap roles. Distinguish shutdown, communication read-back, and physical isolation verification without inventing operating steps.

A short stop message can be respectful

Safety clarification works best when the concern is specific and the pause is unambiguous. Say which instruction is unclear and what needs confirmation. Do not bury the concern in an apology or turn uncertainty into a request to guess the missing step.

F Say it / transfer to a new case

Clarification is not authorization

A classroom card omits the machine identity and approved procedure. No live equipment work has occurred. The trainer will obtain reviewed material.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

missing / exercise / restart / procedure

01 **Learner:** The equipment identity is [1] _____.

02 **Trainer:** We must obtain the applicable approved [2] _____.

03 **Learner:** This discussion remains a classroom [3] _____.

04 **Trainer:** It does not authorize a [4] _____.

CHECK THE GAPS

1. missing / **2.** procedure / **3.** exercise / **4.** restart

Why these answers fit, page 136

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Stop at the unclear instruction". Hana: "Once the correct information is available, I can repeat the equipment identity and the next communication step. That would help confirm that I heard the clarification correctly." Oscar: [missing response] Hana: "The worksheet asks me to mark the machine isolated. I am leaving that blank: we are in a classroom and have not checked any equipment." Which response belongs in the gap?

A. That is a clear stop-work concern. We are in a classroom exercise, and we should not continue as though the missing equipment information were already known.

B. We need the current equipment-specific procedure and the relevant qualified personnel. The correct document and assigned roles matter as much as the words on the card.

C. Yes. Tagout has particular functions and limitations under the applicable system. A warning tag must not be described as providing identical physical restraint to a lock.

D. That is read-back. It checks our shared understanding; it does not replace the required technical process or demonstrate that the equipment has reached a particular state.

Answers and explanations, page 137

Review 3 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Maintenance, Reliability, and Changeover" and "EHS and Safety Communication". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: Work performed to restore function after a fault is identified. Which term fits?

A. corrective maintenance

B. external setup

C. critical asset

D. wear indicator

2. A colleague needs the term for this meaning: Energy that remains after an initial control or shutdown action. Which term fits?

A. residual energy

B. equipment-specific procedure

C. personal protective equipment

D. EHS

3. Rebuild the exchange from "The supplier changed, but so did the process". Rosa: "Can you link each run to the actual material lot? The monthly supplier summary will not tell us whether the old stock was still being used." Jun: [missing response] Rosa: "The sorting team has been screening output while we investigate. The draft calls that corrective action complete, which sounds stronger than the work establishes." Which response belongs in the gap?

A. It does not yet. The supplier is a hypothesis. Two process settings changed in the same period, and the comparable lot records have not been reconciled.

B. Exactly. The observed correlation does not establish which change produced the defect. We need to examine the competing explanations rather than select the easiest one to describe.

C. That is where lot traceability matters. Match the actual lots, settings, and observations before assuming that every unit made after the announcement used the new material.

D. Call it a containment action. Sorting may limit the movement of affected output, but it does not by itself stop the process from producing the defect.

Review 3 / Meaning and context / continued

4. Rebuild the exchange from "Stop at the unclear instruction". Hana: "I will keep the concern specific: missing equipment identity and procedure. The training card needs correction, rather than a handwritten guess about the missing steps." Oscar: [missing response] Hana: "Then this step stays on hold until the reviewed instruction is available. I'll make sure nobody mistakes our clarification for permission to start cleaning." Which response belongs in the gap?

A. That is a clear stop-work concern. We are in a classroom exercise, and we should not continue as though the missing equipment information were already known.

B. We need the current equipment-specific procedure and the relevant qualified personnel. The correct document and assigned roles matter as much as the words on the card.

C. I will refer it through the EHS and training process. We should preserve the unclear wording so the responsible people can review and correct the source.

D. Yes. Tagout has particular functions and limitations under the applicable system. A warning tag must not be described as providing identical physical restraint to a lock.

Answers and explanations, page 154

A Read the situation

LESSON 07 / SUPPLIER QUALITY AND INCOMING MATERIALS

Compare measured results against explicit revision limits and resolve the governing requirement before assigning responsibility or release status.

One dimension, two drawing revisions

Supplier-quality engineer Farah and supplier contact Colin review a component measured at 12.18 mm. The incoming report omits the drawing revision. The inspector used revision C, specifying 12.00 plus or minus 0.10 mm; the supplier used revision B, specifying 12.20 plus or minus 0.10 mm. Which revision governs the order has not been confirmed. The lot is held pending review. Farah and Colin must state both comparisons accurately, trace the order documents, and avoid declaring either party responsible before resolving the requirement.

People: Farah, Supplier-quality engineer; Colin, Supplier quality contact.

Read the record

Choose one answer per question using the facts above.

1. What is the revision C range?

A. 12.10 to 12.30 mm
B. Exactly 12.18 mm only
C. 11.00 to 13.00 mm
D. 11.90 to 12.10 mm

2. How does 12.18 mm compare with revision B's stated range?

A. It lies within 12.10 to 12.30 mm.
B. It exceeds 12.30 mm.
C. It is below 12.10 mm.
D. It determines that revision B governs the order.

3. What remains unconfirmed?

A. Which drawing revision governs this order
B. The reported measured value
C. That the inspector used C
D. That the supplier used B

Answers and explanations, page 138

B Vocabulary in use

CORE TERMS

engineering drawing
A controlled technical representation of product requirements.

consult the engineering drawing

drawing revision
The identified version of a technical drawing.

confirm the drawing revision

nominal dimension
The stated target or reference size.

identify the nominal dimension

tolerance
The permitted variation from a specified value.

apply the stated tolerance

upper specification limit
The largest value permitted by the stated requirement.

check the upper specification limit

lower specification limit
The smallest value permitted by the stated requirement.

check the lower specification limit

measured value
The result reported by the measurement process.

record the measured value

measurement unit
The scale used to express a measurement, such as millimeters.

state the measurement unit

inspection characteristic
A specified feature selected for inspection.

identify the inspection characteristic

inspection report
A record of examined features, methods, and results.

correct the inspection report

purchase order
The commercial document defining the ordered goods and referenced requirements.

review the purchase order

order acknowledgment
The supplier's confirmation of the order terms it received or accepted.

compare the order acknowledgment

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

engineering drawing ENGINEERING DRAWING

consult the engineering drawing

Keep the meaning precise: A controlled technical representation of product requirements.

lower specification limit LOWER SPECIFICATION LIMIT

check the lower specification limit

Keep the meaning precise: The smallest value permitted by the stated requirement.

B Precision and professional shorthand

EXTEND YOUR RANGE

applicable revision

The document version governing the specified work or order.

establish the applicable revision

controlled copy

A document copy maintained under the applicable version-control process.

obtain the controlled copy

engineering change notice

A controlled communication describing an approved technical change.

trace the engineering change notice

supplier flow-down

Communication of applicable requirements through the supply chain.

verify supplier flow-down

lot identity

The identifier linking material to a specific batch.

preserve lot identity

deviation request

A request for authorized departure from a specified requirement.

review a deviation request

document discrepancy

A mismatch between relevant records or references.

resolve a document discrepancy

superseded revision

A version replaced by a later one in the relevant document system.

identify a superseded revision

effectivity

The point or population from which a change applies.

confirm change effectivity

receiving record

Documentation of material received and its identifying details.

trace the receiving record

acceptance basis

The requirements and evidence used for an acceptance decision.

state the acceptance basis

release decision

The authorized determination that material may proceed to use or distribution.

document the release decision

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

order acknowledgment ORDER ACKNOWLEDGMENT

compare the order acknowledgment

Keep the meaning precise: The supplier's confirmation of the order terms it received or accepted.

engineering change notice ENGINEERING CHANGE NOTICE

trace the engineering change notice

Keep the meaning precise: A controlled communication describing an approved technical change.

C Phrases that move the work forward

State the result fully
The reported dimension is 12.18 millimeters.

Show both bases
The inspector used C; the supplier used B.

Read the B limits
Revision B permits 12.10 to 12.30 millimeters.

Avoid premature blame
We have not established which revision governs the order.

Check the change timing
Confirm when revision C became applicable to this order or lot.

Keep the hold visible
The lot remains held pending the required review.

Request the controlled source
Send the referenced drawing and relevant change communication.

Record the resolution
Document the applicable revision and the resulting assessment.

Name the missing reference
The inspection report does not identify the drawing revision.

Read the C limits
Revision C permits 11.90 to 12.10 millimeters.

Report the comparison
The result is within B's stated limits and above C's upper limit.

Trace the agreement
Compare the purchase order with the supplier acknowledgment.

Preserve the original record
Add the corrected reference through the controlled process.

Separate dimension and release
Meeting one dimensional range is not the complete release decision.

Clarify a departure
A deviation request is not already an approved exception.

Close the supplier exchange
Resolve the requirement basis before assigning responsibility or promising release.

Precision in real exchanges

Purpose: Read the C limits
Useful wording: Revision C permits 11.90 to 12.10 millimeters.

Purpose: Read the B limits
Useful wording: Revision B permits 12.10 to 12.30 millimeters.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. What is the difference between 12.18 mm and revision C's upper limit?

A. 0.10 mm below the limit

B. 0.08 mm above the limit

C. 0.18 mm above the limit

D. 0.02 mm below the limit

2. Which update keeps the numerical result separate from the unresolved governing requirement?

A. C is the later letter, so it automatically governed every earlier order.

B. 12.18 meets B's 12.10-12.30 range and exceeds C's upper limit by 0.08; applicability and release remain open.

C. 12.18 meets B, so B must be the applicable requirement and the lot can be released.

D. 12.18 exceeds C's nominal by 0.18, so it exceeds C's upper limit by 0.18.

3. A colleague needs the term for this meaning: The permitted variation from a specified value. Which term fits?

A. tolerance

B. inspection characteristic

C. controlled copy

D. lot identity

D Choose the response that fits / continued

4. Rebuild the exchange from "One dimension, two drawing revisions". Colin: "C says 12.00, plus or minus 0.10. The upper end looks like the lower end of B, which may be why the report became confusing." Farah: [missing response] Colin: "Then neither team needs a different measurement to explain the disagreement. The same 12.18 gets a different comparison because the references differ." Which response belongs in the gap?

A. B's lower specification limit is 12.10 millimeters, and its upper limit is 12.30. The measured 12.18 is inside that interval.

B. C's upper specification limit is 12.10. The measurement is 0.08 above it, with C's full range running from 11.90 to 12.10.

C. Keep the measured value unchanged. We need to resolve the reference, not adjust the recorded number until it fits a preferred drawing.

D. Include the order acknowledgment and attached drawing. I will compare them with our purchase order and the controlled requirement references.

Answers and explanations, page 138

> **One useful distinction**
>
> Revision B permits 12.10 to 12.30 mm; revision C permits 11.90 to 12.10 mm. The measured 12.18 mm is within the supplied B limits and 0.08 mm above C's upper limit. Neither comparison decides which revision governs the order.

E Conversation

One dimension, two drawing revisions

One dimension, two drawing revisions

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

effectivity / supplier flow-down / measured value / order acknowledgment / release decision / upper specification limit / deviation request / acceptance basis / drawing revision / lower specification limit

01 **Colin:** We measured 12.18 millimeters and reported it within tolerance. Your incoming report says it failed. Can we compare the requirement before we debate the result?

02 **Farah:** Yes. The report is missing the [1] _____. Your team used B, our inspector used C, and we have not confirmed which governs this order.

03 **Colin:** B says nominal 12.20, plus or minus 0.10 millimeters. We used that drawing in production, but I agree its applicability still needs checking.

04 **Farah:** B's [2] _____ is 12.10 millimeters, and its upper limit is 12.30. The measured 12.18 is inside that interval.

05 **Colin:** C says 12.00, plus or minus 0.10. The upper end looks like the lower end of B, which may be why the report became confusing.

06 **Farah:** C's [3] _____ is 12.10. The measurement is 0.08 above it, with C's full range running from 11.90 to 12.10.

07 **Colin:** Then neither team needs a different measurement to explain the disagreement. The same 12.18 gets a different comparison because the references differ.

08 **Farah:** Keep the [4] _____ unchanged. We need to resolve the reference, not adjust the recorded number until it fits a preferred drawing.

09 **Colin:** Our production team says B came with the order. I will send the actual package rather than ask you to accept that recollection.

10 **Farah:** Include the [5] _____ and attached drawing. I will compare them with our purchase order and the controlled requirement references.

Conversation continues on the next page.

One dimension, two drawing revisions

CONVERSATION / CONTINUED

11 **Colin:** I found a later change notice too. Its issue date does not tell me whether it applied to material already ordered.

12 **Farah:** Check its [6] _____: the order, lot, date, or other stated application point. Newest in the system does not necessarily mean governing this lot.

13 **Colin:** If C was meant to apply, we also need to know whether the requirement reached our team. I will trace the communication history.

14 **Farah:** That is the [7] _____ question. Keep it evidence-based; a mismatch does not yet tell us which party caused it.

15 **Colin:** Production wants a release estimate. Can I say the lot is usable because 12.18 meets B, while we finish the paperwork?

16 **Farah:** No. The [8] _____ is unresolved and the lot is held. A favorable comparison with B does not supply the missing authorization.

17 **Colin:** We could request a formal exception if one is needed. I will keep that separate from asking which revision actually governed the order.

18 **Farah:** Correct. A [9] _____ is a request, not an approved departure. Its submission would not itself remove the hold.

19 **Colin:** My update will show both ranges, the unchanged measurement, and the documents we are tracing. I will not promise use of the lot.

20 **Farah:** Once the requirement is resolved, record the authorized [10] _____ and correct the report transparently. Preserve the measurement, references, and history of the discrepancy.

CHECK THE GAPS

1. drawing revision / **2.** lower specification limit / **3.** upper specification limit / **4.** measured value / **5.** order acknowledgment / **6.** effectivity / **7.** supplier flow-down / **8.** acceptance basis / **9.** deviation request / **10.** release decision

Why these answers fit, page 139

F Check accuracy and scope

One dimension, two drawing revisions

Return to the case

Supplier-quality engineer Farah and supplier contact Colin review a component measured at 12.18 mm. The incoming report omits the drawing revision. The inspector used revision C, specifying 12.00 plus or minus 0.10 mm; the supplier used revision B, specifying 12.20 plus or minus 0.10 mm. Which revision governs the order has not been confirmed. The lot is held pending review. Farah and Colin must state both comparisons accurately, trace the order documents, and avoid declaring either party responsible before resolving the requirement.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which evidence best addresses the mismatch?

A. A verbal claim that both revisions are equivalent

B. Order references, acknowledgment, drawing versions, and change effectivity records

C. Only the most recent filename in someone's downloads

D. The supplier's production deadline alone

2. How should the report be corrected?

A. Delete the original record and release immediately.

B. Remove units and tolerances to avoid disagreement.

C. Add the missing revision and resolution through the controlled record process while retaining traceability.

D. Replace the result silently so it appears to meet C.

Answers and explanations, page 141

Listen for the distinction

Swap roles. State 0.08 above C's upper limit; preserve the measurement and the unresolved order effectivity.

Resolve the reference before arguing about compliance

Two people can make internally consistent statements while using different documents. Put the requirement identity and revision beside the result, then trace the agreed order. This avoids treating a reference mismatch as dishonesty and keeps the discussion focused on evidence and responsibility.

F Say it / transfer to a new case

Pass against which revision?

A measurement is 5.08 mm. Revision A allows 4.90-5.10 mm; revision B allows 4.95-5.05 mm. The governing revision is unconfirmed and the lot remains held.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

released / unconfirmed / A / limit

01 **Engineer:** The result is within revision [1] _____ limits.

02 **Supplier:** It exceeds revision B's upper [2] _____.

03 **Engineer:** The applicable revision remains [3] _____.

04 **Supplier:** The lot has not been [4] _____.

CHECK THE GAPS

1. A / **2.** limit / **3.** unconfirmed / **4.** released

Why these answers fit, page 141

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "One dimension, two drawing revisions". Colin: "If C was meant to apply, we also need to know whether the requirement reached our team. I will trace the communication history." Farah: [missing response] Colin: "Production wants a release estimate. Can I say the lot is usable because 12.18 meets B, while we finish the paperwork?" Which response belongs in the gap?

A. B's lower specification limit is 12.10 millimeters, and its upper limit is 12.30. The measured 12.18 is inside that interval.

B. That is the supplier flow-down question. Keep it evidence-based; a mismatch does not yet tell us which party caused it.

C. Include the order acknowledgment and attached drawing. I will compare them with our purchase order and the controlled requirement references.

D. C's upper specification limit is 12.10. The measurement is 0.08 above it, with C's full range running from 11.90 to 12.10.

Answers and explanations, page 142

A Read the situation

LESSON 08 / SHIFT HANDOFFS AND ESCALATION

Transfer unfinished work with specific identifiers, evidence, ownership, dependencies, and a confirmed next update.

Six checks complete, two still open

Outgoing supervisor Diego hands lot L27 to incoming supervisor Elena. Six of eight required release checks are complete, with records attached. Check 7, label-count reconciliation, remains open; check 8, final quality review, depends on it. Site procedure requires all eight before lot release. Elena is the named incoming supervisor but has not yet confirmed receipt. A draft handoff says 75% ready and should be fine. Diego and Elena must replace that reassurance with a precise status and a closed-loop handoff.

People: Diego, Outgoing supervisor; Elena, Incoming supervisor.

Read the record

Choose one answer per question using the facts above.

1. Which checks remain open?

A. Check 7 label-count reconciliation and check 8 final quality review
B. All eight checks
C. None because six are complete
D. Only an unspecified optional check

2. What is required before lot release?

A. Merely naming the incoming supervisor
B. Completion of all eight required checks under the site procedure
C. A 75% completion percentage alone
D. A friendly verbal reassurance

3. What has Elena not yet confirmed at the start?

A. That Diego is on the outgoing shift
B. That the lot is called L27
C. That eight checks exist
D. Receipt of the handoff

Answers and explanations, page 143

B Vocabulary in use

CORE TERMS

shift handoff
Transfer of relevant work status and responsibility between shifts.

conduct a shift handoff

outgoing shift
The team ending its scheduled work period.

brief the outgoing shift

incoming shift
The team beginning responsibility for the next period.

inform the incoming shift

open item
A task or issue that has not met its completion criteria.

identify open items

completed check
A required check with its completion evidence recorded.

verify completed checks

release checklist
The defined checks required for a material or product release decision.

review the release checklist

label reconciliation
Accounting for issued, used, unused, and rejected labels as required.

complete label reconciliation

final quality review
The designated review of the required evidence before a release decision.

complete the final quality review

dependency
A task or condition that must be resolved before another can proceed.

identify a task dependency

handoff record
The documented status and responsibilities transferred between teams.

update the handoff record

attachment
A supporting file or record linked to the handoff.

verify the attachment

record reference
An identifier locating the supporting evidence.

provide the record reference

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

shift handoff SHIFT HANDOFF

conduct a shift handoff

Keep the meaning precise: Transfer of relevant work status and responsibility between shifts.

release checklist RELEASE CHECKLIST

review the release checklist

Keep the meaning precise: The defined checks required for a material or product release decision.

B Precision and professional shorthand

EXTEND YOUR RANGE

named owner

A specifically identified person responsible for an action.

assign a named owner

closed-loop communication

An exchange in which receipt and understanding are confirmed.

use closed-loop communication

next checkpoint

The agreed time or event for the next status review.

set the next checkpoint

escalation trigger

A condition requiring referral to the next responsible level.

define an escalation trigger

status qualifier

A word or phrase defining the limits of a reported status.

preserve status qualifiers

continuity

Preservation of information and control across changes of personnel.

maintain continuity

acknowledgment

Confirmation that a message was received, distinct from accepting responsibility.

obtain acknowledgment

read-back

Repetition of key information to verify shared understanding.

request a read-back

due time

The time by which a task is expected or required to be complete.

distinguish due time from update time

handover acceptance

Confirmation of responsibility for the transferred follow-up.

record handover acceptance

pending decision

A required determination that has not yet been made.

flag a pending decision

audit trail

A traceable history of records, actions, and decisions.

preserve the audit trail

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

record reference RECORD REFERENCE

provide the record reference

Keep the meaning precise: An identifier locating the supporting evidence.

next checkpoint NEXT CHECKPOINT

set the next checkpoint

Keep the meaning precise: The agreed time or event for the next status review.

C Phrases that move the work forward

Identify the lot

This handoff concerns lot L27.

State completed work

Six required checks are complete, with the records attached.

Name each open item

Check 7 is label reconciliation; check 8 is final quality review.

Describe the dependency

The final review depends on completion of the reconciliation.

Keep release explicit

The lot is not released while the required checks remain open.

Replace vague reassurance

Please remove should be fine from the status.

Name the recipient

Elena is the incoming supervisor responsible for follow-up.

Confirm receipt

Please confirm you can access the attached records.

Request a precise read-back

Repeat the two open checks and the release restriction.

Separate ownership

Accepting the handoff does not mean the checks are complete.

Set an update

Agree the next checkpoint without promising an unverified finish time.

Define escalation

If required evidence is unavailable, follow the site's escalation route.

Preserve references

Keep the record identifiers beside the completed checks.

Avoid a misleading percentage

Seventy-five percent counts checks; it does not authorize release.

Record changes

Update each status when supporting evidence becomes available.

Close the loop

Confirm the handoff, owner, open items, and next communication.

Precision in real exchanges

Purpose: Describe the dependency
Useful wording: The final review depends on completion of the reconciliation.

Purpose: Keep release explicit
Useful wording: The lot is not released while the required checks remain open.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which handoff line is most useful?

A. L27: checks 1-6 complete; 7 reconciliation and 8 final review open; lot not released.

B. L27 is 75% ready and should be fine.

C. All checks complete because the incoming supervisor is named.

D. L27 released subject to two required checks that have not occurred.

2. Elena confirms receipt and explicitly accepts follow-up responsibility. What may the record state?

A. Handoff received and ownership accepted; checks 7 and 8 remain open and the lot is not released.

B. Receipt confirmed, so both remaining checks are technically complete.

C. Ownership accepted, so final quality review can precede reconciliation.

D. Handoff accepted, so release is authorized subject to later evidence.

3. A colleague needs the term for this meaning: A task or issue that has not met its completion criteria. Which term fits?

A. status qualifier

B. closed-loop communication

C. dependency

D. open item

D Choose the response that fits / continued

4. Rebuild the exchange from "Six checks complete, two still open". Diego: "The completed-check records are attached, with identifiers next to checks one through six. I have named you as the incoming supervisor, but I still need confirmation." Elena: [missing response] Diego: "Thank you. The site procedure requires all eight checks before release, so the two unfinished checks mean L27 has not been released." Which response belongs in the gap?

A. Please replace that with the actual open item list. A percentage by check count does not tell me which work remains or whether the lot can be released.

B. Nine will be the next checkpoint, not a guaranteed finish time. I will report the actual status and any unresolved issue rather than make completion a condition for communicating.

C. First, let me verify each attachment is accessible. A file listed in a message is not useful evidence if the incoming team cannot open it or identify what it supports.

D. It also preserves the audit trail. Later updates can show when each check was completed and who made the decision, without changing what was actually open at this handoff.

Answers and explanations, page 143

> **One useful distinction**
>
> Six of eight checks is 75% by count, but it is not 75% release authorization or a measure of remaining risk. The supplied procedure requires all eight. Name the two open checks and keep the lot-release restriction explicit.

E Conversation

Six checks complete, two still open

Six checks complete, two still open

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

audit trail / handover acceptance / status qualifier / escalation trigger / open item / attachment / dependency / next checkpoint / read-back / release checklist

01 **Diego:** This handoff covers lot L27. Six of the eight required checks are complete, but the draft says seventy-five percent ready and should be fine.

02 **Elena:** Please replace that with the actual [1] _____ list. A percentage by check count does not tell me which work remains or whether the lot can be released.

03 **Diego:** Check seven is label-count reconciliation. Check eight is the final quality review, and that review depends on the completed reconciliation.

04 **Elena:** I will preserve that [2] _____. Assigning both tasks does not make them independent or allow the final review to assume a result that is still missing.

05 **Diego:** The completed-check records are attached, with identifiers next to checks one through six. I have named you as the incoming supervisor, but I still need confirmation.

06 **Elena:** First, let me verify each [3] _____ is accessible. A file listed in a message is not useful evidence if the incoming team cannot open it or identify what it supports.

07 **Diego:** Thank you. The site procedure requires all eight checks before release, so the two unfinished checks mean L27 has not been released.

08 **Elena:** That restriction belongs beside the [4] _____. Six completed items do not reduce the requirement to finish the other two, regardless of how reassuring the summary sounds.

09 **Diego:** I will delete should be fine. What exact status would you want the incoming team to see first?

10 **Elena:** Use a clear [5] _____ instead: not released, two checks open. That tells the team what the status means without suggesting a result we do not have.

Conversation continues on the next page.

Six checks complete, two still open

CONVERSATION / CONTINUED

11 **Diego:** Can you confirm the records and the two remaining checks now? I want the handoff to be more than a message sent just before I leave.

12 **Elena:** Yes. Here is my [6] _____: L27, checks one through six complete; seven reconciliation and eight final review open; final review depends on seven; no release yet.

13 **Diego:** That matches. You are taking responsibility for coordinating the follow-up, but the record should not mark either check complete just because you accepted the handoff.

14 **Elena:** Correct. Record [7] _____ separately from check completion. I accept the follow-up responsibility; the actual check results and authorized decision still need their own evidence.

15 **Diego:** Can we agree on an update at nine tomorrow, even if the reconciliation is still open? I would rather hear the actual position than wait for good news.

16 **Elena:** Nine will be the [8] _____, not a guaranteed finish time. I will report the actual status and any unresolved issue rather than make completion a condition for communicating.

17 **Diego:** If a required record is unavailable, we need the team to use the site's escalation process instead of treating silence as permission to move the lot.

18 **Elena:** I will state that [9] _____ in the handoff. The appropriate route remains the site process; I will not invent an alternative approval because a record is delayed.

19 **Diego:** The corrected record now names the lot, evidence, two open checks, dependency, release restriction, owner, and update time. That is clearer than a readiness percentage.

20 **Elena:** It also preserves the [10] _____. Later updates can show when each check was completed and who made the decision, without changing what was actually open at this handoff.

CHECK THE GAPS

1. open item / **2.** dependency / **3.** attachment / **4.** release checklist / **5.** status qualifier / **6.** read-back / **7.** handover acceptance / **8.** next checkpoint / **9.** escalation trigger / **10.** audit trail

Why these answers fit, page 144

F Check accuracy and scope

Six checks complete, two still open

Return to the case

Outgoing supervisor Diego hands lot L27 to incoming supervisor Elena. Six of eight required release checks are complete, with records attached. Check 7, label-count reconciliation, remains open; check 8, final quality review, depends on it. Site procedure requires all eight before lot release. Elena is the named incoming supervisor but has not yet confirmed receipt. A draft handoff says 75% ready and should be fine. Diego and Elena must replace that reassurance with a precise status and a closed-loop handoff.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. The teams agree an update at 09:00. Which interpretation is safest and most precise?

A. A status report is due at 09:00; completion is not guaranteed unless separately established.

B. Both checks must already be complete because an update time exists.

C. The lot is released at 09:00 regardless of results.

D. No communication is needed unless everything finishes.

2. Which task order matches the brief?

A. Complete label reconciliation before the dependent final quality review.

B. Complete final review by assuming reconciliation will pass later.

C. Mark both complete when the attachments are opened.

D. Remove check 7 because six others are finished.

Answers and explanations, page 146

Listen for the distinction

Swap roles. Confirm receipt and accept responsibility explicitly; make nine tomorrow an update time, not a completion promise.

Ownership is not completion

An incoming supervisor can accept responsibility for follow-up without declaring the work finished. A good read-back identifies the open tasks and restrictions, not just the phrase got it. This matters especially when a friendly handoff risks turning an optimistic expectation into an apparent release decision.

F Say it / transfer to a new case

Acknowledged is not completed

Lot M8 has three of four required checks complete. The last review is open. Noor accepts follow-up responsibility and agrees to provide an update at 16:00, with no completion guarantee.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

time / responsibility / open / release

01 **Outgoing lead:** The final review remains [1] _____.

02 **Noor:** I accept follow-up [2] _____.

03 **Outgoing lead:** Sixteen hundred is the agreed update [3] _____.

04 **Noor:** The handoff does not itself authorize [4] _____.

CHECK THE GAPS

1. open / **2.** responsibility / **3.** time / **4.** release

Why these answers fit, page 146

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Six checks complete, two still open". Diego: "That matches. You are taking responsibility for coordinating the follow-up, but the record should not mark either check complete just because you accepted the handoff." Elena: [missing response] Diego: "Can we agree on an update at nine tomorrow, even if the reconciliation is still open? I would rather hear the actual position than wait for good news." Which response belongs in the gap?

A. That restriction belongs beside the release checklist. Six completed items do not reduce the requirement to finish the other two, regardless of how reassuring the summary sounds.

B. Correct. Record handover acceptance separately from check completion. I accept the follow-up responsibility; the actual check results and authorized decision still need their own evidence.

C. I will preserve that dependency. Assigning both tasks does not make them independent or allow the final review to assume a result that is still missing.

D. Yes. Here is my read-back: L27, checks one through six complete; seven reconciliation and eight final review open; final review depends on seven; no release yet.

Answers and explanations, page 147

Review 4 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Supplier Quality and Incoming Materials" and "Shift Handoffs and Escalation". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: The stated target or reference size. Which term fits?

A. receiving record
B. measurement unit
C. nominal dimension
D. document discrepancy

2. A colleague needs the term for this meaning: The designated review of the required evidence before a release decision. Which term fits?

A. shift handoff
B. named owner
C. final quality review
D. escalation trigger

3. Rebuild the exchange from "Stop at the unclear instruction". Hana: "I understand the need to ask, but I do not want my question to sound as though I am volunteering to perform the isolation myself." Oscar: [missing response] Hana: "Some learners also treat a tag as though it physically prevents the same actions as a lock. We should make the distinction clear without demonstrating an unapproved method." Which response belongs in the gap?

A. You are not. The authorized employee has an assigned role and the required training. Participating in this discussion does not give you that authorization.
B. Yes. Tagout has particular functions and limitations under the applicable system. A warning tag must not be described as providing identical physical restraint to a lock.
C. That is a clear stop-work concern. We are in a classroom exercise, and we should not continue as though the missing equipment information were already known.
D. We need the current equipment-specific procedure and the relevant qualified personnel. The correct document and assigned roles matter as much as the words on the card.

Review 4 / Meaning and context / continued

4. Rebuild the exchange from "Six checks complete, two still open". Diego: "If a required record is unavailable, we need the team to use the site's escalation process instead of treating silence as permission to move the lot." Elena: [missing response] Diego: "The corrected record now names the lot, evidence, two open checks, dependency, release restriction, owner, and update time. That is clearer than a readiness percentage." Which response belongs in the gap?

A. Use a clear status qualifier instead: not released, two checks open. That tells the team what the status means without suggesting a result we do not have.

B. Please replace that with the actual open item list. A percentage by check count does not tell me which work remains or whether the lot can be released.

C. That restriction belongs beside the release checklist. Six completed items do not reduce the requirement to finish the other two, regardless of how reassuring the summary sounds.

D. I will state that escalation trigger in the handoff. The appropriate route remains the site process; I will not invent an alternative approval because a record is delayed.

Answers and explanations, page 157

Language for the next case

The gauge passes today, but what about yesterday?

as-found results

As-found results describe the instrument before adjustment, which is the relevant starting point for assessing earlier measurements.

Marta: Not yet. I need the as-found results, not just the passing report after adjustment. What did it read before anyone changed it?

as-left results

As-left results describe the instrument after the calibration work or adjustment and do not establish its earlier condition.

Marta: Thank you. The as-left results tell us its condition afterward. They do not resolve the earlier acceptance decisions.

asset identifier

The asset identifier connects the specific instrument to its inspection records, avoiding assumptions based on a tool description alone.

Marta: Match those sheets to the instrument's asset identifier. We must not include every micrometer just because they look alike.

measurement uncertainty

Measurement uncertainty describes the dispersion associated with a measured value; it contributes to evaluating whether the measurement is fit for its intended decision.

Marta: The measured features, limits, method, and relevant measurement uncertainty. Then we can assess which decisions may be affected.

reinspection

Reinspection is a new examination under an established plan; its results must remain distinguishable from the original inspection.

Marta: Correct. Preserve the original records. Any reinspection needs an agreed plan, suitable equipment, and separate results.

fitness for purpose

Fitness for purpose means the measurement capability is adequate for the intended task, not merely that a calibration label exists.

Marta: Check its range, resolution, condition, and suitability for the feature. A calibration label alone does not establish fitness for purpose.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

The gauge passes today, but what about yesterday?

A micrometer fails its incoming calibration check and passes after adjustment. It was used to accept three production lots yesterday. The quality engineer and supervisor must organize a retrospective review. The fictional site has removed the instrument from production use pending the required decisions.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

as-found results / measurement uncertainty / asset identifier / reinspection / fitness for purpose / as-left results

01 **Cal:** The lab adjusted the micrometer and it passes now. Can I close the issue and put yesterday's three lots back on schedule?

02 **Marta:** Not yet. I need the [1] _____, not just the passing report after adjustment. What did it read before anyone changed it?

03 **Cal:** I have the final certificate here. The earlier readings are on a separate sheet; I will request that too.

04 **Marta:** Thank you. The [2] _____ tell us its condition afterward. They do not resolve the earlier acceptance decisions.

05 **Cal:** We know it was used yesterday, but two operators shared it. Their inspection sheets should give us the part numbers.

06 **Marta:** Match those sheets to the instrument's [3] _____. We must not include every micrometer just because they look alike.

07 **Cal:** Should I tell the warehouse that all three lots are defective? They are asking what to put in their update.

08 **Marta:** Say acceptance is under review. A suspect measurement does not automatically prove that every part is defective.

09 **Cal:** What else will you need once we have the readings and the lot list?

10 **Marta:** The measured features, limits, method, and relevant [4] _____. Then we can assess which decisions may be affected.

Conversation continues on the next page.

The gauge passes today, but what about yesterday?

CONVERSATION / CONTINUED

11 **Cal:** Some results were near a limit. I will flag those, but I will not quietly replace them with fresh measurements.

12 **Marta:** Correct. Preserve the original records. Any [5] _____ needs an agreed plan, suitable equipment, and separate results.

13 **Cal:** One of the lots has already left the site. I will identify its destination rather than leave it out of the review.

14 **Marta:** Please do. Include shipped material in the impact assessment and route any required customer action to the responsible team.

15 **Cal:** Can we borrow another calibrated micrometer for the next job while you review these lots?

16 **Marta:** Check its range, resolution, condition, and suitability for the feature. A calibration label alone does not establish [6] _____.

17 **Cal:** Understood. I will send the three lot records and identify which instrument handled each inspection before the planning meeting.

18 **Marta:** I will bring the initial assessment and list anything still missing. Keep the affected material status visible until the required disposition is recorded.

19 **Cal:** The update will separate today's instrument condition from yesterday's product decisions. That should prevent the passing certificate from closing both questions.

20 **Marta:** Exactly. We need a defensible conclusion about the earlier measurements, not just a tool that reads correctly now.

CHECK THE GAPS

1. as-found results / **2.** as-left results / **3.** asset identifier / **4.** measurement uncertainty / **5.** reinspection / **6.** fitness for purpose

Why these answers fit, page 160

A replacement gauge is not automatically suitable

A replacement instrument has a current calibration label, but its range and measurement capability have not been checked against the required feature.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

measurement / limits / inspection / instrument

01 **Supervisor:** The label does not establish fitness for this [1] _____.

02 **Engineer:** Compare the instrument capability with the feature and its [2] _____.

03 **Supervisor:** Keep earlier results separate from any new [3] _____.

04 **Engineer:** Record the basis for selecting the replacement [4] _____.

CHECK THE GAPS

1. measurement / **2.** limits / **3.** inspection / **4.** instrument

Why these answers fit, page 160

Keep the distinction audible

Report a measurement concern and define the scope of a product-impact review.

Language for the next case

The right total, the wrong replenishment mix

part-number mix

The part-number mix describes the quantities of different variants; the correct combined total does not establish the required mix.

Dev: I see the problem. The part-number mix is sixty left-hand and twenty right-hand, rather than forty of each.

pick list

The pick list records the items and quantities requested for retrieval, helping locate where the variant mismatch entered the process.

Dev: I will check the pick list against what was actually loaded. We should not assume the error started in stores.

inventory balance

The inventory balance must reflect the actual quantity and location after the physical movement, rather than retaining an incorrect delivery total.

Dev: No. I will return the excess through the normal process and correct the inventory balance after the movement is recorded.

shortage

The shortage is twenty right-hand brackets relative to the forty required; excess left-hand brackets do not substitute for them.

Dev: So the immediate shortage is twenty right-hand brackets. I will locate those before asking stores for another eighty parts.

kanban

A kanban is a signal authorizing production or movement within a defined pull system; it must identify the relevant item and quantity.

Dev: Agreed. I will match each kanban to its part number and standard quantity before changing anything.

Held stock

Held stock is restricted pending the applicable decision; physical availability does not mean it is available for production.

Dev: Leave it alone. Held stock is not a source we can use just to clear the shortage.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

The right total, the wrong replenishment mix

An assembly cell needs 40 left-hand and 40 right-hand brackets. Its delivery contains 60 left-hand and 20 right-hand brackets. The total of 80 is correct, but only 20 complete pairs are available. A material handler and cell leader reconcile the delivery and the replenishment signal.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

shortage / inventory balance / kanban / pick list / Held stock / part-number mix

01 **Inez:** Dev, the trolley has eighty brackets, but we cannot build forty assemblies from it. Can you check the two variant labels with me?

02 **Dev:** I see the problem. The [1] _____ is sixty left-hand and twenty right-hand, rather than forty of each.

03 **Inez:** Each assembly uses one of each. That gives us twenty complete pairs, with forty left-hand brackets left over.

04 **Dev:** So the immediate [2] _____ is twenty right-hand brackets. I will locate those before asking stores for another eighty parts.

05 **Inez:** Thank you. The last delivery note only showed the family total. We need the two part numbers on the correction.

06 **Dev:** I will check the [3] _____ against what was actually loaded. We should not assume the error started in stores.

07 **Inez:** The line-side bins look similar, and one replenishment card was tucked behind the other. That may be relevant, but let us preserve the actual record.

08 **Dev:** Agreed. I will match each [4] _____ to its part number and standard quantity before changing anything.

09 **Inez:** Please do not increase both quantities as a quick fix. We have limited space, and extra left-hand stock will not solve the missing pairs.

10 **Dev:** No. I will return the excess through the normal process and correct the [5] _____ after the movement is recorded.

Conversation continues on the next page.

The right total, the wrong replenishment mix

CONVERSATION / CONTINUED

11 **Inez:** Can you give me a reliable arrival time for the twenty missing pieces? I need to coordinate the remaining work at the cell.

12 **Dev:** I will confirm stock and transport first. At the moment, I can promise a status update in ten minutes, not delivery in ten minutes.

13 **Inez:** That works. There is also a right-hand container on the next trolley, but it carries a hold label.

14 **Dev:** Leave it alone. [6] _____ is not a source we can use just to clear the shortage.

15 **Inez:** Once the correct parts arrive, we will verify the identifiers and quantities at the cell, not only the total on the trolley.

16 **Dev:** I will include those details on the delivery record. Will you confirm receipt so stores knows the correction reached the right location?

17 **Inez:** Yes. Then we can review whether the card position or the family-only list contributed to the mismatch.

18 **Dev:** I will bring both records. Let us fix the immediate shortage first and investigate the signal without guessing who made the mistake.

19 **Inez:** Good. I will tell planning we have twenty pairs now and twenty more pending, rather than eighty usable components for forty assemblies.

20 **Dev:** That describes the actual constraint. I will come back with the stock position and the confirmed delivery arrangement.

CHECK THE GAPS

1. part-number mix / **2.** shortage / **3.** pick list / **4.** kanban / **5.** inventory balance / **6.** Held stock

Why these answers fit, page 162

A family total hides a variant shortage

A cell needs 12 large and 12 small housings. A delivery contains 18 large and six small housings. Each assembly requires one of each.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

housings / record / assemblies / receipt

01 **Leader:** We can build only six complete [1] _____.

02 **Handler:** We are short of six small [2] _____.

03 **Leader:** Reconcile the two variants separately on the delivery [3] _____.

04 **Handler:** Verify the corrected quantities when the cell confirms [4] _____.

CHECK THE GAPS

1. assemblies / **2.** housings / **3.** record / **4.** receipt

Why these answers fit, page 162

Keep the distinction audible

Resolve a line-side material shortage without confusing quantities, variants, and replenishment signals.

Language for the next case

The first-piece report points to the wrong feature

ballooned drawing

A ballooned drawing numbers the features to connect them with inspection results; the number must refer to the correct feature and revision.

Rae: On this ballooned drawing, feature twelve is the ten-millimeter slot. Let us check whether we are discussing the same feature.

inspection characteristic

The inspection characteristic is the specific feature or requirement being evaluated, not merely a number copied between documents.

Rae: Revision D. The inspection characteristic needs to match the actual slot, including its applicable requirements.

controlled correction

A controlled correction preserves the original record and documents the authorized change, rather than silently replacing an inconsistent result.

Rae: Please keep the original entry. We need a controlled correction, not a number changed to make the failure disappear.

datum

A datum is the theoretically exact reference used for applicable dimensional or geometric requirements; the reference must match the drawing and method.

Rae: Exactly. We will reconcile the references and inspect the correct feature using the approved method. The datum reference matters where the requirement calls for it.

First-piece acceptance

First-piece acceptance is the recorded determination that the required setup checks meet their criteria; making or submitting the piece does not establish acceptance.

Rae: That should say first piece submitted. First-piece acceptance has not been recorded, and the unresolved inspection cannot be hidden by the schedule label.

in-process checks

In-process checks monitor the subsequent production according to the established plan; first-piece acceptance does not remove those requirements.

Rae: No. The in-process checks remain part of the production plan. A satisfactory first piece does not prove the whole run will remain satisfactory.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

The first-piece report points to the wrong feature

A machine setup produces a first piece for review. The report calls a 6 mm hole feature 12, but feature 12 on the current ballooned drawing is a 10 mm slot. The setup technician and inspector need to reconcile the report, drawing, and physical part. No production approval has been recorded.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

First-piece acceptance / inspection characteristic / datum / ballooned drawing / controlled correction / in-process checks

01 **Nico:** The first piece is ready, but your report marks feature twelve as failed. I measured the six-millimeter hole before bringing it over.

02 **Rae:** On this [1] _____, feature twelve is the ten-millimeter slot. Let us check whether we are discussing the same feature.

03 **Nico:** That explains the mismatch. My setup sheet calls the hole twelve. Which drawing revision does your report identify?

04 **Rae:** Revision D. The [2] _____ needs to match the actual slot, including its applicable requirements.

05 **Nico:** My sheet also says D, but the feature numbering may have been copied from an earlier report. I will bring the source file.

06 **Rae:** Please keep the original entry. We need a [3] _____, not a number changed to make the failure disappear.

Conversation / continued

07 **Nico:** Agreed. I also want to know whether the part itself meets the requirement. Fixing the label will not answer that.

08 **Rae:** Exactly. We will reconcile the references and inspect the correct feature using the approved method. The [4] _____ reference matters where the requirement calls for it.

09 **Nico:** Would a passing result for the hole still be useful, provided it is connected to the right characteristic?

10 **Rae:** Yes, if the evidence supports it. We should not discard a valid observation, but we cannot relabel it as a measurement of the slot.

Conversation continues on the next page.

The first-piece report points to the wrong feature

CONVERSATION / CONTINUED

11 **Nico:** Planning is asking whether the machine can start the full run. They see first piece complete on the schedule.

12 **Rae:** That should say first piece submitted. [5] _____ has not been recorded, and the unresolved inspection cannot be hidden by the schedule label.

13 **Nico:** I will correct that status now. Is this also the same as approving every part that follows?

14 **Rae:** No. The [6] _____ remain part of the production plan. A satisfactory first piece does not prove the whole run will remain satisfactory.

15 **Nico:** We should check the other copied feature numbers too. Otherwise this may be only the first mismatch we have noticed.

16 **Rae:** I agree. Let us compare the full mapping before closing the report. I will identify which results are affected and which remain correctly linked.

17 **Nico:** I can provide the setup sheet and the earlier template. You can show the actual inspection records beside the current drawing.

18 **Rae:** Good. Then the reviewer can follow the correction instead of reconstructing it from a verbal explanation later.

19 **Nico:** I will tell planning the setup is awaiting inspection resolution, with no production approval yet. We will give them the next status after review.

20 **Rae:** And we will separate the documentation mismatch from the part result. Either may need action, but they are not the same finding.

CHECK THE GAPS

1. ballooned drawing / **2.** inspection characteristic / **3.** controlled correction / **4.** datum / **5.** First-piece acceptance / **6.** in-process checks

Why these answers fit, page 164

A submitted first piece is shown as accepted

The setup team submits a first piece, but a required dimensional result is missing and the acceptance decision is still open.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

feature / run / accepted / approval

01 **Technician:** The piece is submitted, but it is not yet [1] _____.

02 **Inspector:** Obtain the missing result for the specified [2] _____.

03 **Technician:** Keep the schedule status separate from production [3] _____.

04 **Inspector:** Retain the required checks throughout the subsequent [4] _____.

CHECK THE GAPS

1. accepted / **2.** feature / **3.** approval / **4.** run

Why these answers fit, page 164

Keep the distinction audible

Resolve a drawing-to-inspection mismatch before treating a setup as approved.

01 / Answers and explanations

A / Read the situation

A1. What was the output shortfall?

D. 160 units

Subtracting 840 completed units from the 1,000-unit target gives a 160-unit shortfall.

Return to the exercise, page 4

A2. What does 84% represent here?

A. Output as a share of the stated target

The available calculation is 840 divided by 1,000, which measures output target attainment.

Return to the exercise, page 4

A3. What is known about the downtime?

C. Forty minutes were logged; its cause and contribution remain under review.

The briefing gives a duration but not the cause or a complete loss attribution.

Return to the exercise, page 4

D / Response choices

D1. Which board entry is accurate?

D. 840 units completed; 84% of target; 160-unit shortfall.

The entry uses only the supplied output and target and keeps units distinct from time.

Return to the exercise, page 8

D2. Why is OEE not established?

B. Planned time, ideal cycle time, and quality information are not supplied.

OEE combines distinct factors whose required inputs are missing from the brief.

Return to the exercise, page 8

01 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: The amount by which a result falls below its target. Which term fits?

D. shortfall

A Not here. work in process: Material that has entered production but is not yet finished. The usual combination is "control work in process".

B Not here. unplanned stop: An interruption not included in the intended operating schedule. The usual combination is "investigate unplanned stops".

C Not here. tier meeting: A short escalation meeting at a defined organizational level. The usual combination is "escalate through a tier meeting".

D Correct. shortfall: The amount by which a result falls below its target. The usual combination is "quantify the shortfall".

Return to the exercise, page 8

D4. Rebuild the exchange from "Eight hundred forty is not the whole story". Daniel: "The stop log records forty minutes. Does the team believe that interruption accounts for all the missing units, or is the analysis still open?" Mei: [missing response] Daniel: "We would need to know the expected production rate and whether other losses occurred. The board currently makes a single interruption look like a complete explanation." Which response belongs in the gap?

C. The downtime is logged, but its cause and contribution remain under review. We should not convert forty minutes into one hundred sixty lost units without the relevant basis.

A Not here. This line belongs to turn 4, after Daniel says: "Please change the heading before the morning meeting. What is the gap in units, and what can we actually say about the stop?" It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 18, after Daniel says: "For today's meeting, I need a concise update: result, known interruption, open questions, and responsibility for the next check. We can discuss recovery after that evidence is clearer." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Mei's response in turn 6. It connects Daniel's preceding point with Daniel's follow-up. Downtime is the recorded nonproducing duration, whose cause and contribution remain unconfirmed here.

D Not here. This line belongs to turn 8, after Daniel says: "We would need to know the expected production rate and whether other losses occurred. The board currently makes a single interruption look like a complete explanation." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 8

01 / Answers and explanations / continued

E / Conversation

1. target attainment

Turn 2. The calculation compares actual output with the target, not the separate inputs needed for OEE.

2. shortfall

Turn 4. The shortfall is the target minus actual output: 1,000 minus 840 equals 160 units.

3. downtime

Turn 6. Downtime is the recorded nonproducing duration, whose cause and contribution remain unconfirmed here.

4. Loss attribution

Turn 8. Loss attribution assigns losses to supported contributors; the log alone does not complete that analysis.

5. availability

Turn 10. Availability relates running time to planned production time, which is not supplied in this brief.

6. performance loss

Turn 12. Performance loss concerns running below the defined ideal rate, requiring speed-related information beyond downtime.

7. Quality loss

Turn 14. Quality loss concerns units not meeting the defined good-product criteria, information absent from the supplied count.

8. bottleneck

Turn 16. A bottleneck is the system-limiting step, which must be identified from evidence rather than a local impression.

9. visual board

Turn 18. The visual board should distinguish the confirmed measures, unresolved causes, and assigned next actions.

10. recovery plan

Turn 20. A recovery plan is a proposal for addressing the gap, not proof that the missing output will be recovered.

Return to the conversation, page 10

01 / Answers and explanations / continued

F / Accuracy and scope

F1. A manager converts the 40-minute stop into the full 160-unit shortfall. What is missing from that attribution?

A. A supported running-rate and loss breakdown linking the stop to the output gap

The count gap is known, but minutes cannot become lost units without a relevant rate and loss basis. Other speed, quality, or operating losses may also contribute.

Return to the exercise, page 12

F2. If a second shift produces 900 against a target of 1,000, what is its attainment?

A. 90%, six percentage points above 84%

900 divided by 1,000 is 90%; the difference between 90% and 84% is six percentage points.

Return to the exercise, page 12

F / Guided speaking

1. ninety

Lead: Target attainment is ninety percent.

Turn 1. Dividing 450 by 500 gives 0.90, or ninety percent.

2. fifty

Planner: The output shortfall is fifty units.

Turn 2. The target of 500 minus actual output of 450 leaves fifty units.

3. downtime

Lead: Twenty minutes describes recorded downtime.

Turn 3. The supplied duration is nonproducing time, not a quantity of missing units.

4. unconfirmed

Planner: The cause remains unconfirmed.

Turn 4. The briefing provides no established explanation for the recorded stop.

Return to the conversation, page 13

01 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Eight hundred forty is not the whole story". Daniel: "And the completed-unit count does not tell us how many met the quality definition. We should not silently assume that every produced unit was good." Mei: [missing response] Daniel: "Someone has suggested the packing station is the limiting step. That may be worth checking, but I have not seen evidence showing it constrained the whole line." Which response belongs in the gap?

D. Correct. Quality loss needs the appropriate output breakdown. A production count by itself cannot establish that no defects or other quality losses occurred.

A Not here. This line belongs to turn 2, after Daniel says: "Yesterday's board shows eight hundred forty units against a thousand. It labels eighty-four percent as OEE, but I cannot see the inputs behind that label." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 16, after Daniel says: "Someone has suggested the packing station is the limiting step. That may be worth checking, but I have not seen evidence showing it constrained the whole line." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 10, after Daniel says: "Which inputs do you need for the equipment measure? Give me the missing fields so I can ask the line team for them." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Mei's response in turn 14. It connects Daniel's preceding point with Daniel's follow-up. Quality loss concerns units not meeting the defined good-product criteria, information absent from the supplied count.

Return to the exercise, page 13

02 / Answers and explanations

A / Read the situation

A1. What was observed?

D. Eight tool-waiting episodes totaling 24 operator-minutes

The observation concerns waiting episodes and accumulated operator time, not a completed improvement or individual performance diagnosis.

Return to the exercise, page 14

A2. Why is 24 not necessarily line downtime?

B. Some waiting episodes overlap, and the measure is operator-minutes.

Overlapping individual waits cannot simply be added and labeled elapsed whole-line downtime.

Return to the exercise, page 14

A3. What is the proposed change?

B. A one-bench point-of-use tool trial with safety and quality checks unchanged

The scenario supplies a bounded trial proposal, not a result or permission to remove controls.

Return to the exercise, page 14

D / Response choices

D1. Which observation is defensible?

C. Eight tool-waiting episodes totaled 24 operator-minutes during the hour.

The statement preserves the actual event count, observation period, and accumulated time measure.

Return to the exercise, page 18

D2. What is the average time per recorded episode?

C. Three operator-minutes

Dividing 24 operator-minutes by eight episodes gives three operator-minutes per episode.

Return to the exercise, page 18

02 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Movement of people within a task or process. Which term fits?

C. motion

A Not here. 5S: Workplace organization through sort, set in order, shine, standardize, and sustain. The usual combination is "sustain 5S practices".

B Not here. baseline observation: A measurement of the current process before a proposed change. The usual combination is "record a baseline observation".

C Correct. motion: Movement of people within a task or process. The usual combination is "analyze unnecessary motion".

D Not here. kaizen: Focused, ongoing improvement of work processes. The usual combination is "run a kaizen event".

Return to the exercise, page 18

D4. Rebuild the exchange from "The queue is for a shared tool". Marcus: "I will put the one-hour observation period beside that figure. We did not watch a full shift, so I cannot call it a shift total." Ana: [missing response] Marcus: "The operators suggested a tool location beside the bench. It sounds practical, but we need to understand whether other stations depend on that same shared tool." Which response belongs in the gap?

C. Exactly. This is our baseline observation. Record the task conditions and tool demand so the later comparison is meaningful rather than just another isolated total.

A Not here. This line belongs to turn 16, after Marcus says: "Hold on, the slide already says twenty-four minutes saved. Nobody has run the trial. Can we change that before it reaches the review meeting?" It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 4, after Marcus says: "The sheet adds up to twenty-four minutes. I was going to call that twenty-four minutes of line downtime, although two operators sometimes waited at the same time." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Ana's response in turn 6. It connects Marcus's preceding point with Marcus's follow-up. The baseline records the current process before a change and needs a clear time and task basis.

D Not here. This line belongs to turn 18, after Marcus says: "After the trial, we should ask whether a reduction here created a tool shortage elsewhere. A faster bench would not necessarily mean a better overall flow." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 18

E / Conversation

1. waiting

Turn 2. Waiting describes work delayed by unavailable input, not evidence of a slow working pace.

02 / Answers and explanations / continued

2. operator-minute

Turn 4. An operator-minute measures one person's time; overlapping waits cannot be relabeled as elapsed whole-line downtime.

3. baseline observation

Turn 6. The baseline records the current process before a change and needs a clear time and task basis.

4. current-state map

Turn 8. A current-state map represents the existing locations, activities, and information flow before proposing a changed arrangement.

5. motion

Turn 10. Motion concerns people's movements within work, distinct from moving material between locations.

6. point-of-use storage

Turn 12. Point-of-use storage places needed items near their use, here as a proposal rather than a verified result.

7. pilot trial

Turn 14. A pilot trial is a bounded test whose conditions and retained controls must be clear before interpreting results.

8. countermeasure

Turn 16. A countermeasure is an action intended to address a problem; it does not establish achieved savings before testing.

9. value stream

Turn 18. The value stream includes connected activities, so a local change must be assessed for effects elsewhere.

10. sustainment

Turn 20. Sustainment concerns maintaining an accepted improvement through the agreed method and follow-up rather than a one-time result.

Return to the conversation, page 20

02 / Answers and explanations / continued

F / Accuracy and scope

F1. Which proposal has suitable boundaries?

D. Trial tool placement at one bench, keep required controls, and compare waiting under comparable conditions.

The proposal defines a limited comparison without changing required controls or assuming the outcome.

Return to the exercise, page 23

F2. Waiting falls at the trial bench while the shared tool is less available elsewhere. Which conclusion still needs evidence?

B. Total benefit across the connected process, including any displaced waiting

Moving delay can improve a local measure without improving the whole process. Compare relevant areas on a consistent basis before claiming overall benefit.

Return to the exercise, page 23

F / Guided speaking

1. operator-minutes

Facilitator: The accumulated waiting measure is operator-minutes.

Turn 1. The case adds individuals' waiting time, including overlapping episodes.

2. three

Supervisor: Average waiting per episode is three minutes.

Turn 2. Twelve operator-minutes divided by four episodes gives three per episode.

3. trial

Facilitator: The proposed tool location still needs a trial.

Turn 3. No evaluation of the proposed location has yet been performed.

4. savings

Supervisor: We cannot report achieved savings yet.

Turn 4. A proposal has no measured improvement result before it is tested.

Return to the conversation, page 24

02 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "The queue is for a shared tool". Marcus: "We can start with one bench and compare the waiting pattern. Required inspection and safety checks should stay in place, so the trial does not gain time by removing controls." Ana: [missing response] Marcus: "Hold on, the slide already says twenty-four minutes saved. Nobody has run the trial. Can we change that before it reaches the review meeting?" Which response belongs in the gap?

B. Make that explicit in the pilot trial description. Use comparable task conditions and the same measure, and record any new difficulties rather than reporting only the favorable observations.

A Not here. This line belongs to turn 12, after Marcus says: "For the trial, the suggestion is to keep an appropriate tool at this one bench. We have not decided that every work area needs a new tool arrangement." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Ana's response in turn 14. It connects Marcus's preceding point with Marcus's follow-up. A pilot trial is a bounded test whose conditions and retained controls must be clear before interpreting results.

C Not here. This line belongs to turn 2, after Marcus says: "We watched the bench for an hour and recorded eight delays. The team says the operators are slow, but each delay seemed to involve waiting for the shared tool." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 10, after Marcus says: "That would also help us separate people walking to retrieve the tool from material being moved between work areas. Both happen, but they are different kinds of movement." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 24

03 / Answers and explanations

A / Read the situation

A1. What is the first-pass yield?

C. 90%

Nine hundred first-pass accepted units divided by 1,000 inspected units gives ninety percent.

Return to the exercise, page 27

A2. How many units have approved scrap disposition?

D. 40

Only forty units are in the supplied scrap category; sixty await rework.

Return to the exercise, page 27

A3. How many reworked units are already accepted?

B. None

The scenario explicitly states that no reworked units have yet been accepted.

Return to the exercise, page 27

D / Response choices

D1. Which report reconciles the batch correctly?

D. 900 first-pass accepted, 60 awaiting rework, and 40 scrap

The three supplied mutually exclusive states total 1,000 without counting pending recovery as accepted.

Return to the exercise, page 31

D2. If all 60 later pass the required acceptance process, what would the accepted total be?

D. 960, while first-pass yield remains 90%

Later acceptance can increase final accepted quantity without changing how many passed on their first attempt.

Return to the exercise, page 31

03 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Product designated not to be used for its original intended purpose. Which term fits?

B. scrap

A Not here. rework queue: Units waiting for the approved rework process. The usual combination is "monitor the rework queue".

B Correct. scrap: Product designated not to be used for its original intended purpose. The usual combination is "record approved scrap".

C Not here. rework labor: Labor used to bring product into conformity through rework. The usual combination is "record rework labor".

D Not here. mutually exclusive categories: Groups defined so that an item belongs to only one of them. The usual combination is "use mutually exclusive categories".

Return to the exercise, page 31

D4. Rebuild the exchange from "One thousand units, three different states". Ellis: "The dashboard also shows ninety-six percent first-pass yield. Someone added the sixty units they expect to recover, even though none has passed after rework yet." Priya: [missing response] Ellis: "Then the current accepted count is nine hundred. I can show a possible nine hundred sixty separately, but not as output already available." Which response belongs in the gap?

D. The first-pass yield is ninety percent. Nine hundred passed at first inspection. A later recovery does not change that original first-pass result.

A Not here. This line belongs to turn 14, after Ellis says: "When the work finishes, should I move the units straight into accepted output? The work-order status will tell me that the operation is complete." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 12, after Ellis says: "The technician called the work repair in an email. Should I use that word in the report, or retain the process named in the record?" It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 8, after Ellis says: "Then the current accepted count is nine hundred. I can show a possible nine hundred sixty separately, but not as output already available." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Priya's response in turn 6. It connects Ellis's preceding point with Ellis's follow-up. First-pass yield is 900 divided by 1,000, excluding any later rework recovery.

Return to the exercise, page 31

E / Conversation

1. scrap

Turn 2. Scrap is the forty-unit disposition category, not all one hundred units that failed to pass initially.

03 / Answers and explanations / continued

2. mutually exclusive categories

Turn 4. Exclusive categories place each unit in one state, allowing the total to reconcile without overlap.

3. first-pass yield

Turn 6. First-pass yield is 900 divided by 1,000, excluding any later rework recovery.

4. rework queue

Turn 8. The rework queue contains units awaiting the process, not units already recovered and accepted.

5. recovery assumption

Turn 10. A recovery assumption describes a possible future accepted quantity and must not be reported as an achieved result.

6. rework

Turn 12. Rework aims to restore conformity to requirements, whereas repair can have a different acceptance basis.

7. reinspection

Turn 14. Reinspection provides the specified post-work inspection evidence and is not replaced by completion of the operation alone.

8. defect count

Turn 16. Defect count can exceed defective-unit count because one unit may contain several distinct defects.

9. double-counting

Turn 18. Double-counting would include recovered units in both the pending and accepted states instead of transferring their status.

10. material reconciliation

Turn 20. Material reconciliation accounts for the batch across its actual states before quantities are used in cost or production summaries.

Return to the conversation, page 33

03 / Answers and explanations / continued

F / Accuracy and scope

F1. One unit has three defects and is included in the sixty awaiting rework. How should the unit reconciliation treat it?

B. One pending unit, with three defects recorded in the separate defect count

The physical item appears once in its current unit category. The three defects are a different measure; neither a work plan nor multiple defects changes its acceptance or scrap status.

Return to the exercise, page 35

F2. Which financial statement is supported without further data?

D. Scrap quantity and pending rework are known; their complete costs are not supplied.

The case supplies quantities but no complete costing basis for scrap or rework.

Return to the exercise, page 35

F / Guided speaking

1. eighty-five

Analyst: First-pass yield is eighty-five percent.

Turn 1. 170 divided by 200 is 0.85, or eighty-five percent.

2. twenty

Manager: The pending rework quantity is twenty.

Turn 2. Twenty units await rework and are not yet accepted recovered output.

3. ten

Analyst: Approved scrap is ten units.

Turn 3. The scrap category contains ten units, separate from pending rework.

4. conditional

Manager: A total of 190 accepted units remains conditional.

Turn 4. Reaching 190 requires all twenty pending units to pass the required acceptance process.

Return to the conversation, page 36

03 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "One thousand units, three different states". Ellis: "When the work finishes, should I move the units straight into accepted output? The work-order status will tell me that the operation is complete." Priya: [missing response] Ellis: "Another table lists three defects on one unit. I need to keep that from becoming three defective units in the batch reconciliation." Which response belongs in the gap?

B. Wait for the required reinspection and acceptance decision. A finished operation does not by itself establish that the units meet requirements.

A Not here. This line belongs to turn 18, after Ellis says: "And when a unit is accepted after rework, it leaves the pending category. Otherwise our total would grow without any additional material." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Priya's response in turn 14. It connects Ellis's preceding point with Ellis's follow-up. Reinspection provides the specified post-work inspection evidence and is not replaced by completion of the operation alone.

C Not here. This line belongs to turn 8, after Ellis says: "Then the current accepted count is nine hundred. I can show a possible nine hundred sixty separately, but not as output already available." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 12, after Ellis says: "The technician called the work repair in an email. Should I use that word in the report, or retain the process named in the record?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 36

04 / Answers and explanations

A / Read the situation

A1. What changed in the relevant period?

B. The supplier and two process settings

Multiple changes occurred, so the timeline does not isolate the supplier as the cause.

Return to the exercise, page 37

A2. What evidence is still unreconciled?

A. Comparable lot and process-setting records

The brief says the relevant lot and setting records have not yet been reconciled.

Return to the exercise, page 37

A3. What is the status of temporary sorting?

D. An immediate screening measure, not proof of recurrence prevention

Sorting can identify affected output while leaving the underlying cause unresolved.

Return to the exercise, page 37

D / Response choices

D1. Which root-cause statement fits the evidence?

D. The supplier change and two setting changes remain competing explanations pending comparison.

Multiple simultaneous changes leave the cause unresolved but identify evidence that can help examine the alternatives.

Return to the exercise, page 41

D2. What does sorting establish by itself?

D. A screening activity is being used; recurrence prevention is not demonstrated.

Sorting addresses detection or containment and does not by itself change the process producing the defect.

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04 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: An observed association that does not by itself establish causation. Which term fits?

D. correlation

A Not here. sorting: Separating items using defined inspection or classification criteria. The usual combination is "document temporary sorting".

B Not here. comparative evidence: Information allowing relevant conditions or groups to be compared. The usual combination is "obtain comparative evidence".

C Not here. recurrence: The same or a related problem appearing again. The usual combination is "monitor recurrence".

D Correct. correlation: An observed association that does not by itself establish causation. The usual combination is "distinguish correlation from causation".

Return to the exercise, page 41

D4. Rebuild the exchange from "The supplier changed, but so did the process". Rosa: "The process settings could affect the result directly or interact with the supplied material. The current summary treats them as background details instead of relevant changes." Jun: [missing response] Rosa: "What should the evidence package contain? The identifiers in our defect, receiving, and machine records do not yet line up." Which response belongs in the gap?

D. Each is a possible confounding factor. If conditions changed together, we cannot attribute the outcome to the supplier alone without further supporting comparison.

A Not here. This line belongs to turn 10, after Rosa says: "Can you link each run to the actual material lot? The monthly supplier summary will not tell us whether the old stock was still being used." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 16, after Rosa says: "That leaves a second investigation line: why the previous check did not flag the issue. Solving that could improve detection even before the production explanation is settled." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Rosa says: "The draft says the new supplier is the root cause because the defect first appeared after the switch. I want to check whether the evidence really isolates that change." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Jun's response in turn 6. It connects Rosa's preceding point with Rosa's follow-up. A confounding factor can influence the apparent relationship between the supplier switch and the defect.

Return to the exercise, page 41

E / Conversation

1. hypothesis

Turn 2. The supplier is a possible explanation awaiting evidence, not a confirmed cause established by sequence.

04 / Answers and explanations / continued

2. correlation

Turn 4. Correlation describes association and does not isolate causation when other relevant variables also changed.

3. confounding factor

Turn 6. A confounding factor can influence the apparent relationship between the supplier switch and the defect.

4. change history

Turn 8. The change history establishes which modifications occurred and when, allowing records to be compared accurately.

5. lot traceability

Turn 10. Lot traceability connects specific material lots with production and defect records instead of relying on broad period labels.

6. containment action

Turn 12. Containment limits impact while the cause is investigated; sorting alone does not demonstrate recurrence prevention.

7. occurrence cause

Turn 14. The occurrence cause explains why the defect was produced, a question separate from whether sorting detects it.

8. detection failure

Turn 16. A detection failure explains why a check missed the problem and may require an action different from the production cause.

9. effectiveness check

Turn 18. An effectiveness check evaluates whether the action achieved the intended prevention result rather than merely being documented.

10. closure criteria

Turn 20. Closure criteria specify the evidence required before the issue can be accepted as resolved.

Return to the conversation, page 43

04 / Answers and explanations / continued

F / Accuracy and scope

F1. Which request best tests the competing supplier and setting explanations?

D. Link defect observations to actual supplier lots and both settings under comparable conditions.

Records linked to each relevant factor permit meaningful comparisons. Period labels, voting, and containment completion do not isolate a cause when several inputs changed.

Return to the exercise, page 46

F2. What distinguishes an occurrence cause from a detection failure?

A. One explains production of the defect; the other explains why a check missed it.

Producing a defect and failing to detect it are distinct questions that may require different actions.

Return to the exercise, page 46

F / Guided speaking

1. hypothesis

Engineer: The material explanation is still a hypothesis.

Turn 1. Multiple changes occurred, so the material has not been established as the cause.

2. factor

Reviewer: The setting change is a potential confounding factor.

Turn 2. The other changing variable can affect the apparent material-defect relationship.

3. action

Engineer: Temporary sorting is a containment action.

Turn 3. Screening limits impact while the cause remains under investigation.

4. evidence

Reviewer: Recurrence prevention still needs effectiveness evidence.

Turn 4. Active containment does not establish that a corrective action prevents the defect from recurring.

Return to the conversation, page 47

04 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "The supplier changed, but so did the process". Rosa: "We should still evaluate the sorting method through the appropriate procedure. A temporary measure should not be treated as perfect simply because it is currently in place." Jun: [missing response] Rosa: "That leaves a second investigation line: why the previous check did not flag the issue. Solving that could improve detection even before the production explanation is settled." Which response belongs in the gap?

D. Agreed. And keep the occurrence cause separate from the screening question. One concerns why the defect is created; the other concerns whether the existing controls find it.

A Not here. This line belongs to turn 4, after Rosa says: "I will keep the dates and remove the confirmed-cause label. Which competing explanations do you want the investigation team to check?" It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 8, after Rosa says: "What should the evidence package contain? The identifiers in our defect, receiving, and machine records do not yet line up." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Rosa says: "The draft says the new supplier is the root cause because the defect first appeared after the switch. I want to check whether the evidence really isolates that change." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Jun's response in turn 14. It connects Rosa's preceding point with Rosa's follow-up. The occurrence cause explains why the defect was produced, a question separate from whether sorting detects it.

Return to the exercise, page 47

05 / Answers and explanations

A / Read the situation

A1. When do the two demands overlap?

B. From 13:00 to 14:00

Maintenance starts at 13:00 while the current rush-order schedule needs the line until 14:00.

Return to the exercise, page 50

A2. What is known about maintenance deferral?

D. No supporting technical assessment or authorization is supplied.

The brief leaves the assessment and decision open rather than approving a schedule change.

Return to the exercise, page 50

A3. What is the alternative-line status?

D. Possible but unconfirmed in suitability

A potential option does not establish the line's technical suitability, readiness, or allocation.

Return to the exercise, page 50

D / Response choices

D1. Which schedule statement is accurate?

A. There is a one-hour overlap, but the maintenance scope remains two hours unless properly reassessed.

The overlap calculation does not shorten the planned work or authorize a schedule change.

Return to the exercise, page 54

D2. Which alternative-line statement is best?

B. Assess suitability, setup, readiness, and allocation before committing the order.

A credible contingency needs more than the possibility of unused capacity.

Return to the exercise, page 54

05 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Tracking indicators of equipment health over time. Which term fits?

B. condition monitoring

A Not here. spare-part readiness: Availability of the correct parts needed for the planned work. The usual combination is "confirm spare-part readiness".

B Correct. condition monitoring: Tracking indicators of equipment health over time. The usual combination is "review condition-monitoring data".

C Not here. deferral: Postponement of a planned task subject to the required assessment and decision. The usual combination is "assess a maintenance deferral".

D Not here. reliability: The ability to perform the required function over a stated period under stated conditions. The usual combination is "assess equipment reliability".

Return to the exercise, page 54

D4. Rebuild the exchange from "A rush order meets the maintenance window". Ben: "The customer will ask why we cannot use a machine that ran yesterday. What can I say without suggesting that yesterday's run settles today's maintenance decision?" Leila: [missing response] Ben: "What information should we bring to the decision maker? I can show the order timing and the impact on delivery, but the work scope needs your input." Which response belongs in the gap?

D. It does not establish reliability for the proposed period. The task's purpose, equipment condition, and consequences need assessment rather than a conclusion based only on yesterday's operation.

A Not here. This line belongs to turn 14, after Ben says: "The improvement team has discussed moving preparation outside stopped-machine time. We should not assume every preparation activity can be moved that way." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 8, after Ben says: "What information should we bring to the decision maker? I can show the order timing and the impact on delivery, but the work scope needs your input." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 4, after Ben says: "Could we move the work later? I need an option for the customer discussion, but I do not have the technical basis to say that postponement is acceptable." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Leila's response in turn 6. It connects Ben's preceding point with Ben's follow-up. Reliability concerns performance over defined conditions and time, not a guarantee inferred from one recent operating period.

Return to the exercise, page 54

05 / Answers and explanations / continued

E / Conversation

1. maintenance window

Turn 2. The maintenance window is the reserved period from 13:00 to 15:00, which conflicts with the order schedule.

2. deferral

Turn 4. Deferral is a proposed postponement requiring the relevant assessment and authorization, neither of which is supplied here.

3. reliability

Turn 6. Reliability concerns performance over defined conditions and time, not a guarantee inferred from one recent operating period.

4. work order

Turn 8. The work order identifies the defined maintenance task and its controlled scope for the scheduling assessment.

5. capacity contingency

Turn 10. A capacity contingency is a conditional alternative whose suitability and readiness still need confirmation.

6. changeover

Turn 12. A changeover transitions equipment to another job, while the original task concerns maintenance.

7. External setup

Turn 14. External setup consists of eligible preparation while equipment runs; not every task can safely or appropriately be moved there.

8. spare-part readiness

Turn 16. Spare-part readiness establishes whether the correct needed parts are available for the planned scope.

9. schedule dependency

Turn 18. A schedule dependency identifies a condition that must be resolved before the corresponding plan can be relied on.

05 / Answers and explanations / continued

10. Return to service

Turn 20. Return to service is the authorized restoration of operational use after the relevant work and checks, not merely task completion.

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F / Accuracy and scope

F1. Which reply preserves authority?

B. We can present deferral for assessment, but it is not authorized in the current information.

The supplied case contains no technical basis or authorized deferral decision.

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F2. The maintenance operation is marked finished. What remains a separate status to confirm?

D. Required checks and authorized return to service

Finishing the work does not automatically establish that the required checks and return-to-service decision are complete. The original schedule facts do not replace those controls.

Return to the exercise, page 59

F / Guided speaking

1. one

Planner: The schedules overlap by one hour.

Turn 1. The shared period is 10:00 to 11:00, a one-hour overlap.

2. deferral

Technician: Moving maintenance would be a proposed deferral.

Turn 2. Postponement is not approved merely because an order conflicts with the window.

3. contingency

Planner: The second line is a capacity contingency.

Turn 3. The alternative remains conditional until its suitability and readiness are confirmed.

4. suitability

Technician: Before using it, confirm its suitability.

Turn 4. The briefing specifically leaves the second line's capability for the job unconfirmed.

Return to the conversation, page 60

05 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "A rush order meets the maintenance window". Ben: "The improvement team has discussed moving preparation outside stopped-machine time. We should not assume every preparation activity can be moved that way." Leila: [missing response] Ben: "We also need the correct maintenance parts and people available. A calendar reservation alone does not demonstrate that the work can finish as planned." Which response belongs in the gap?

B. Exactly. External setup refers to eligible tasks that can occur while equipment runs. Required safety, quality, and operating controls still govern how the work is organized.

A Not here. This line belongs to turn 20, after Ben says: "Once the authorized choice is recorded, I will update the affected teams. I will also avoid describing the equipment as ready merely because the work is reported finished." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Leila's response in turn 14. It connects Ben's preceding point with Ben's follow-up. External setup consists of eligible preparation while equipment runs; not every task can safely or appropriately be moved there.

C Not here. This line belongs to turn 6, after Ben says: "The customer will ask why we cannot use a machine that ran yesterday. What can I say without suggesting that yesterday's run settles today's maintenance decision?" It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 4, after Ben says: "Could we move the work later? I need an option for the customer discussion, but I do not have the technical basis to say that postponement is acceptable." It addresses that point, not the missing turn in the supplied exchange.

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06 / Answers and explanations

A / Read the situation

A1. Where is this exchange taking place?

A. In a classroom safety exercise with no equipment work underway

The brief explicitly describes a classroom exercise and supplies no live-work authorization.

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A2. Which clarification is needed before this classroom card could support the intended technical discussion?

B. Identified equipment, energy sources, and the current applicable approved procedure

The card lacks the equipment-specific basis needed to interpret it. A signature or general statement does not supply that basis, and the exercise authorizes no actual work.

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A3. What must not be invented?

A. An isolation sequence for an unidentified machine

The task is to stop and clarify communication, not create a machine procedure without the required basis.

Return to the exercise, page 61

D / Response choices

D1. Which response best fits the ambiguous card?

B. Pause and obtain the current applicable equipment-specific procedure through the authorized route.

The missing context requires clarification by the proper process, not invented operational instructions.

Return to the exercise, page 65

D2. What does switching off establish by itself?

C. Normal operation may be stopped; complete energy isolation is not established.

Shutdown and complete hazardous-energy control are distinct, so the broader conclusion is unsupported.

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06 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Separation from energy sources using the applicable control method. Which term fits?

D. energy isolation

A Not here. safety data sheet: A document describing a chemical's hazards and handling information. The usual combination is "consult the safety data sheet".

B Not here. shutdown: Stopping normal equipment operation. The usual combination is "distinguish shutdown from isolation".

C Not here. stop-work concern: A safety concern requiring work to pause under the applicable process. The usual combination is "raise a stop-work concern".

D Correct. energy isolation: Separation from energy sources using the applicable control method. The usual combination is "follow the energy-isolation procedure".

Return to the exercise, page 65

D4. Rebuild the exchange from "Stop at the unclear instruction". Hana: "There could also be energy remaining after a normal stop. I should not assume that the absence of movement proves there is nothing left that could cause harm." Oscar: [missing response] Hana: "Which machine is this card supposed to cover? Can we get its approved instructions before the class goes any further?" Which response belongs in the gap?

B. Exactly. Stored energy needs the approved process for this equipment. We can't fill the gap by improvising an isolation procedure.

A Not here. This line belongs to turn 2, after Hana says: "I want to stop at this line. The card says switch off, then clean, but it does not identify the machine or the procedure we should be using." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Oscar's response in turn 6. It connects Hana's preceding point with Hana's follow-up. Stored energy can remain after normal supply or operation stops, so it cannot be dismissed from a generic off instruction.

C Not here. This line belongs to turn 10, after Hana says: "I understand the need to ask, but I do not want my question to sound as though I am volunteering to perform the isolation myself." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 16, after Hana says: "The worksheet asks me to mark the machine isolated. I am leaving that blank: we are in a classroom and have not checked any equipment." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 65

E / Conversation

1. stop-work concern

Turn 2. The concern calls for a pause at the ambiguous instruction rather than guessing a work sequence.

06 / Answers and explanations / continued

2. Shutdown

Turn 4. Shutdown stops normal operation but does not establish that every hazardous energy source has been controlled.

3. Stored energy

Turn 6. Stored energy can remain after normal supply or operation stops, so it cannot be dismissed from a generic off instruction.

4. equipment-specific procedure

Turn 8. The equipment-specific procedure supplies the approved instructions for identified equipment rather than a generic classroom shortcut.

5. authorized employee

Turn 10. The authorized employee has the assigned role and required training for the relevant energy-control work.

6. Tagout

Turn 12. Tagout uses warning tags within the applicable system and must not be assumed to provide the same physical restraint as a lock.

7. read-back

Turn 14. Read-back checks shared understanding of critical information but does not establish a physical equipment condition.

8. Verification of isolation

Turn 16. Verification of isolation requires the approved technical method, which has not occurred in this classroom discussion.

9. EHS

Turn 18. EHS identifies the environment, health, and safety function involved in addressing the safety-related training issue.

10. restart authorization

Turn 20. Restart authorization is a separate required permission and is not created by completing a classroom clarification.

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06 / Answers and explanations / continued

F / Accuracy and scope

F1. What can read-back confirm?

B. Shared understanding of the clarified message, not the physical isolated state

Repeating information checks communication and does not replace technical verification or role requirements.

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F2. Which closing statement is accurate?

D. The ambiguous classroom card is referred for correction; no equipment work or restart is authorized.

The exercise identifies a communication problem without supplying live-work or restart authorization.

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F / Guided speaking

1. missing

Learner: The equipment identity is missing.

Turn 1. The card omits the specific machine, so its identity cannot be assumed.

2. procedure

Trainer: We must obtain the applicable approved procedure.

Turn 2. A generic card cannot replace the required equipment-specific instructions.

3. exercise

Learner: This discussion remains a classroom exercise.

Turn 3. The case supplies no actual equipment work or servicing authorization.

4. restart

Trainer: It does not authorize a restart.

Turn 4. A training clarification does not provide permission to resume equipment operation.

Return to the conversation, page 70

06 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Stop at the unclear instruction". Hana: "Once the correct information is available, I can repeat the equipment identity and the next communication step. That would help confirm that I heard the clarification correctly." Oscar: [missing response] Hana: "The worksheet asks me to mark the machine isolated. I am leaving that blank: we are in a classroom and have not checked any equipment." Which response belongs in the gap?

D. That is read-back. It checks our shared understanding; it does not replace the required technical process or demonstrate that the equipment has reached a particular state.

A Not here. This line belongs to turn 2, after Hana says: "I want to stop at this line. The card says switch off, then clean, but it does not identify the machine or the procedure we should be using." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 8, after Hana says: "Which machine is this card supposed to cover? Can we get its approved instructions before the class goes any further?" It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 12, after Hana says: "Some learners also treat a tag as though it physically prevents the same actions as a lock. We should make the distinction clear without demonstrating an unapproved method." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Oscar's response in turn 14. It connects Hana's preceding point with Hana's follow-up. Read-back checks shared understanding of critical information but does not establish a physical equipment condition.

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07 / Answers and explanations

A / Read the situation

A1. What is the revision C range?

D. 11.90 to 12.10 mm

A nominal 12.00 mm with a tolerance of 0.10 gives limits of 11.90 and 12.10.

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A2. How does 12.18 mm compare with revision B's stated range?

A. It lies within 12.10 to 12.30 mm.

The measurement is within the stated B limits, but that comparison does not establish the governing revision.

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A3. What remains unconfirmed?

A. Which drawing revision governs this order

The brief supplies both references but not the agreed revision applicable to the order.

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D / Response choices

D1. What is the difference between 12.18 mm and revision C's upper limit?

B. 0.08 mm above the limit

The upper limit is 12.10 mm, and 12.18 minus 12.10 equals 0.08 mm.

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D2. Which update keeps the numerical result separate from the unresolved governing requirement?

B. 12.18 meets B's 12.10-12.30 range and exceeds C's upper limit by 0.08; applicability and release remain open.

C's upper limit is 12.10, making the excess 0.08, not the distance from nominal. Neither numerical fit nor revision chronology establishes applicability or release authority.

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07 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: The permitted variation from a specified value. Which term fits?

A. tolerance

A Correct. tolerance: The permitted variation from a specified value. The usual combination is "apply the stated tolerance".

B Not here. inspection characteristic: A specified feature selected for inspection. The usual combination is "identify the inspection characteristic".

C Not here. controlled copy: A document copy maintained under the applicable version-control process. The usual combination is "obtain the controlled copy".

D Not here. lot identity: The identifier linking material to a specific batch. The usual combination is "preserve lot identity".

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D4. Rebuild the exchange from "One dimension, two drawing revisions". Colin: "C says 12.00, plus or minus 0.10. The upper end looks like the lower end of B, which may be why the report became confusing." Farah: [missing response] Colin: "Then neither team needs a different measurement to explain the disagreement. The same 12.18 gets a different comparison because the references differ." Which response belongs in the gap?

B. C's upper specification limit is 12.10. The measurement is 0.08 above it, with C's full range running from 11.90 to 12.10.

A Not here. This line belongs to turn 4, after Colin says: "B says nominal 12.20, plus or minus 0.10 millimeters. We used that drawing in production, but I agree its applicability still needs checking." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Farah's response in turn 6. It connects Colin's preceding point with Colin's follow-up. Adding 0.10 to the C nominal 12.00 gives the upper permitted value of 12.10 mm.

C Not here. This line belongs to turn 8, after Colin says: "Then neither team needs a different measurement to explain the disagreement. The same 12.18 gets a different comparison because the references differ." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 10, after Colin says: "Our production team says B came with the order. I will send the actual package rather than ask you to accept that recollection." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 77

E / Conversation

1. drawing revision

Turn 2. The drawing revision identifies the requirement version needed to interpret the inspection result.

07 / Answers and explanations / continued

2. lower specification limit

Turn 4. Subtracting 0.10 from the B nominal 12.20 gives the lower permitted value of 12.10 mm.

3. upper specification limit

Turn 6. Adding 0.10 to the C nominal 12.00 gives the upper permitted value of 12.10 mm.

4. measured value

Turn 8. The measured value remains 12.18 mm; differing reference limits change its interpretation, not the recorded number.

5. order acknowledgment

Turn 10. The acknowledgment records the supplier's received or accepted terms and can be compared with the purchase order references.

6. effectivity

Turn 12. Effectivity specifies when or to which population the technical change applies, resolving more than simple revision chronology.

7. supplier flow-down

Turn 14. Supplier flow-down concerns how applicable requirements were communicated through the supply chain.

8. acceptance basis

Turn 16. The acceptance basis remains unresolved while the applicable revision is unknown, so a favorable B comparison cannot settle release.

9. deviation request

Turn 18. A deviation request seeks an authorized departure and is not itself approval of that departure.

10. release decision

Turn 20. The release decision is the authorized result of the review, not an assumption drawn from one favorable comparison.

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07 / Answers and explanations / continued

F / Accuracy and scope

F1. Which evidence best addresses the mismatch?

B. Order references, acknowledgment, drawing versions, and change effectivity records

The relevant records connect the agreed requirement and change timing to the particular order and lot.

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F2. How should the report be corrected?

C. Add the missing revision and resolution through the controlled record process while retaining traceability.

The correction must preserve the measured result and traceable requirement basis rather than obscure the discrepancy.

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F / Guided speaking

1. A

Engineer: The result is within revision A limits.

Turn 1. 5.08 lies between the supplied A limits of 4.90 and 5.10.

2. limit

Supplier: It exceeds revision B's upper limit.

Turn 2. 5.08 is above the B upper permitted value of 5.05.

3. unconfirmed

Engineer: The applicable revision remains unconfirmed.

Turn 3. The case does not establish which reference governs the order.

4. released

Supplier: The lot has not been released.

Turn 4. The material remains held, so neither comparison supplies release authorization.

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07 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "One dimension, two drawing revisions". Colin: "If C was meant to apply, we also need to know whether the requirement reached our team. I will trace the communication history." Farah: [missing response] Colin: "Production wants a release estimate. Can I say the lot is usable because 12.18 meets B, while we finish the paperwork?" Which response belongs in the gap?

B. That is the supplier flow-down question. Keep it evidence-based; a mismatch does not yet tell us which party caused it.

A Not here. This line belongs to turn 4, after Colin says: "B says nominal 12.20, plus or minus 0.10 millimeters. We used that drawing in production, but I agree its applicability still needs checking." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Farah's response in turn 14. It connects Colin's preceding point with Colin's follow-up. Supplier flow-down concerns how applicable requirements were communicated through the supply chain.

C Not here. This line belongs to turn 10, after Colin says: "Our production team says B came with the order. I will send the actual package rather than ask you to accept that recollection." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 6, after Colin says: "C says 12.00, plus or minus 0.10. The upper end looks like the lower end of B, which may be why the report became confusing." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 82

08 / Answers and explanations

A / Read the situation

A1. Which checks remain open?

A. Check 7 label-count reconciliation and check 8 final quality review

The brief identifies two specific required checks and their dependency.

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A2. What is required before lot release?

B. Completion of all eight required checks under the site procedure

The supplied site procedure requires every release check, not a partial completion threshold.

Return to the exercise, page 83

A3. What has Elena not yet confirmed at the start?

D. Receipt of the handoff

A named recipient is not proof that the information has been received and understood.

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D / Response choices

D1. Which handoff line is most useful?

A. L27: checks 1-6 complete; 7 reconciliation and 8 final review open; lot not released.

The line identifies the actual completed and open work and preserves the stated release restriction.

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D2. Elena confirms receipt and explicitly accepts follow-up responsibility. What may the record state?

A. Handoff received and ownership accepted; checks 7 and 8 remain open and the lot is not released.

Receipt and explicit acceptance establish communication and ownership. They do not complete the checks, remove their dependency, or authorize the lot's release.

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08 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: A task or issue that has not met its completion criteria. Which term fits?

D. open item

A Not here. status qualifier: A word or phrase defining the limits of a reported status. The usual combination is "preserve status qualifiers".

B Not here. closed-loop communication: An exchange in which receipt and understanding are confirmed. The usual combination is "use closed-loop communication".

C Not here. dependency: A task or condition that must be resolved before another can proceed. The usual combination is "identify a task dependency".

D Correct. open item: A task or issue that has not met its completion criteria. The usual combination is "identify open items".

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D4. Rebuild the exchange from "Six checks complete, two still open". Diego: "The completed-check records are attached, with identifiers next to checks one through six. I have named you as the incoming supervisor, but I still need confirmation." Elena: [missing response] Diego: "Thank you. The site procedure requires all eight checks before release, so the two unfinished checks mean L27 has not been released." Which response belongs in the gap?

C. First, let me verify each attachment is accessible. A file listed in a message is not useful evidence if the incoming team cannot open it or identify what it supports.

A Not here. This line belongs to turn 2, after Diego says: "This handoff covers lot L27. Six of the eight required checks are complete, but the draft says seventy-five percent ready and should be fine." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 16, after Diego says: "Can we agree on an update at nine tomorrow, even if the reconciliation is still open? I would rather hear the actual position than wait for good news." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Elena's response in turn 6. It connects Diego's preceding point with Diego's follow-up. Attachments are the supporting records whose availability must be confirmed, not assumed from a sent message.

D Not here. This line belongs to turn 20, after Diego says: "The corrected record now names the lot, evidence, two open checks, dependency, release restriction, owner, and update time. That is clearer than a readiness percentage." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 87

E / Conversation

1. open item

Turn 2. An open item is unfinished work that must be named, rather than hidden behind a readiness percentage.

08 / Answers and explanations / continued

2. dependency

Turn 4. The dependency requires label reconciliation to be completed before the final quality review can be completed.

3. attachment

Turn 6. Attachments are the supporting records whose availability must be confirmed, not assumed from a sent message.

4. release checklist

Turn 8. The release checklist defines the required checks, all of which remain necessary before the stated release decision.

5. status qualifier

Turn 10. A status qualifier such as not released states the limits of the current condition rather than offering vague reassurance.

6. read-back

Turn 12. Read-back repeats the critical lot status, open checks, and restriction to confirm shared understanding.

7. handover acceptance

Turn 14. Handover acceptance confirms responsibility for follow-up and does not establish technical completion of the remaining checks.

8. next checkpoint

Turn 16. The next checkpoint is an agreed status-review time, not a guaranteed completion or release time.

9. escalation trigger

Turn 18. The trigger identifies the missing evidence condition that requires referral through the site's responsible process.

10. audit trail

Turn 20. The audit trail records who knew and accepted what, alongside later evidence and decisions, without rewriting earlier uncertainty.

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08 / Answers and explanations / continued

F / Accuracy and scope

F1. The teams agree an update at 09:00. Which interpretation is safest and most precise?

A. A status report is due at 09:00; completion is not guaranteed unless separately established.

An agreed checkpoint concerns reporting status and should not be turned into an unsupported completion promise.

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F2. Which task order matches the brief?

A. Complete label reconciliation before the dependent final quality review.

The brief explicitly makes final quality review dependent on the completed reconciliation.

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F / Guided speaking

1. open

Outgoing lead: The final review remains open.

Turn 1. Three completed checks leave the fourth review unfinished.

2. responsibility

Noor: I accept follow-up responsibility.

Turn 2. Noor accepts ownership of the follow-up rather than declaring the review complete.

3. time

Outgoing lead: Sixteen hundred is the agreed update time.

Turn 3. The checkpoint is a reporting time, not a guaranteed completion time.

4. release

Noor: The handoff does not itself authorize release.

Turn 4. An accepted handoff cannot replace the remaining required review or release decision.

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08 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Six checks complete, two still open". Diego: "That matches. You are taking responsibility for coordinating the follow-up, but the record should not mark either check complete just because you accepted the handoff." Elena: [missing response] Diego: "Can we agree on an update at nine tomorrow, even if the reconciliation is still open? I would rather hear the actual position than wait for good news." Which response belongs in the gap?

B. Correct. Record handover acceptance separately from check completion. I accept the follow-up responsibility; the actual check results and authorized decision still need their own evidence.

A Not here. This line belongs to turn 8, after Diego says: "Thank you. The site procedure requires all eight checks before release, so the two unfinished checks mean L27 has not been released." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Elena's response in turn 14. It connects Diego's preceding point with Diego's follow-up. Handover acceptance confirms responsibility for follow-up and does not establish technical completion of the remaining checks.

C Not here. This line belongs to turn 4, after Diego says: "Check seven is label-count reconciliation. Check eight is the final quality review, and that review depends on the completed reconciliation." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 12, after Diego says: "Can you confirm the records and the two remaining checks now? I want the handoff to be more than a message sent just before I leave." It addresses that point, not the missing turn in the supplied exchange.

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Review 1 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: Actual output expressed relative to the stated target. Which term fits?

C. target attainment

A Not here. lead time: The elapsed time from a defined request or start to completion. The usual combination is "reduce lead time".

B Not here. planned stop: A scheduled interruption included in the operating plan. The usual combination is "distinguish planned stops".

C Correct. target attainment: Actual output expressed relative to the stated target. The usual combination is "calculate target attainment".

D Not here. daily management: Routine review of performance, problems, and follow-up actions. The usual combination is "support daily management".

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2. A colleague needs the term for this meaning: Material held beyond the justified operating need. Which term fits?

D. excess inventory

A Not here. lean manufacturing: An approach to improving customer value by reducing waste in the process. The usual combination is "apply lean manufacturing".

B Not here. future-state map: A proposed representation of an improved process. The usual combination is "develop a future-state map".

C Not here. work-cell layout: The arrangement of people, equipment, and material in a production cell. The usual combination is "assess the work-cell layout".

D Correct. excess inventory: Material held beyond the justified operating need. The usual combination is "reduce excess inventory".

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Review 1 / Explanations / continued

3. Rebuild the exchange from "Eight hundred forty is not the whole story". Daniel: "Which inputs do you need for the equipment measure? Give me the missing fields so I can ask the line team for them." Mei: [missing response] Daniel: "The speed side also needs a clear reference. Actual output can fall short even while the equipment is running, so time and quantity should not be merged." Which response belongs in the gap?

C. Start with the basis for availability. We need the planned production time and the relevant running time, not merely the output target and one stop duration.

A Not here. This line belongs to turn 8, after Daniel says: "We would need to know the expected production rate and whether other losses occurred. The board currently makes a single interruption look like a complete explanation." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 4, after Daniel says: "Please change the heading before the morning meeting. What is the gap in units, and what can we actually say about the stop?" It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Mei's response in turn 10. It connects Daniel's preceding point with Daniel's follow-up. Availability relates running time to planned production time, which is not supplied in this brief.

D Not here. This line belongs to turn 20, after Daniel says: "Good. We can acknowledge the missed target without guessing why it happened. Any proposal to recover output should make its assumptions and constraints clear." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 25

Review 1 / Explanations / continued

4. Rebuild the exchange from "The queue is for a shared tool". Marcus: "After the trial, we should ask whether a reduction here created a tool shortage elsewhere. A faster bench would not necessarily mean a better overall flow." Ana: [missing response] Marcus: "If the trial supports the change, the team still needs an approved method and follow-up. Otherwise the tool could gradually return to the old shared location." Which response belongs in the gap?

C. Review the broader value stream. Local gains matter, but a change that simply transfers waiting to another step may not improve delivery to the customer.

A Not here. This line belongs to turn 6, after Marcus says: "I will put the one-hour observation period beside that figure. We did not watch a full shift, so I cannot call it a shift total." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 4, after Marcus says: "The sheet adds up to twenty-four minutes. I was going to call that twenty-four minutes of line downtime, although two operators sometimes waited at the same time." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Ana's response in turn 18. It connects Marcus's preceding point with Marcus's follow-up. The value stream includes connected activities, so a local change must be assessed for effects elsewhere.

D Not here. This line belongs to turn 16, after Marcus says: "Hold on, the slide already says twenty-four minutes saved. Nobody has run the trial. Can we change that before it reaches the review meeting?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 25

Review 2 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: A unit that does not meet a specified requirement. Which term fits?

D. nonconforming unit

A Not here. final yield: The share meeting the stated acceptance criteria after the defined processing stages. The usual combination is "state the final-yield basis".

B Not here. material reconciliation: Accounting for all material quantities across defined states. The usual combination is "perform material reconciliation".

C Not here. scrap cost: The defined cost assigned to scrapped product or material. The usual combination is "calculate scrap cost".

D Correct. nonconforming unit: A unit that does not meet a specified requirement. The usual combination is "identify nonconforming units".

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2. A colleague needs the term for this meaning: The ability to link material and records to particular lots. Which term fits?

B. lot traceability

A Not here. root cause: An underlying cause whose removal addresses the problem's recurrence. The usual combination is "establish the root cause".

B Correct. lot traceability: The ability to link material and records to particular lots. The usual combination is "maintain lot traceability".

C Not here. 8D: An eight-discipline structured approach to team problem solving. The usual combination is "prepare an 8D report".

D Not here. corrective action: Action addressing a cause to prevent recurrence. The usual combination is "implement corrective action".

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Review 2 / Explanations / continued

3. Rebuild the exchange from "The queue is for a shared tool". Marcus: "That would also help us separate people walking to retrieve the tool from material being moved between work areas. Both happen, but they are different kinds of movement." Ana: [missing response] Marcus: "For the trial, the suggestion is to keep an appropriate tool at this one bench. We have not decided that every work area needs a new tool arrangement." Which response belongs in the gap?

C. Yes. Unnecessary motion concerns the people performing the task. We should describe the actual movement rather than use a broad waste label that hides what needs to change.

A Not here. This line belongs to turn 12, after Marcus says: "For the trial, the suggestion is to keep an appropriate tool at this one bench. We have not decided that every work area needs a new tool arrangement." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 14, after Marcus says: "We can start with one bench and compare the waiting pattern. Required inspection and safety checks should stay in place, so the trial does not gain time by removing controls." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Ana's response in turn 10. It connects Marcus's preceding point with Marcus's follow-up. Motion concerns people's movements within work, distinct from moving material between locations.

D Not here. This line belongs to turn 2, after Marcus says: "We watched the bench for an hour and recorded eight delays. The team says the operators are slow, but each delay seemed to involve waiting for the shared tool." It addresses that point, not the missing turn in the supplied exchange.

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Review 2 / Explanations / continued

4. Rebuild the exchange from "The supplier changed, but so did the process". Rosa: "Once the evidence supports an action, we need a defined way to judge whether it worked. A completed supplier response form will not be enough." Jun: [missing response] Rosa: "I will revise the status: containment active, supplier and setting hypotheses open, and record reconciliation assigned to you. We will not announce a cause before the evidence supports it." Which response belongs in the gap?

B. Set an effectiveness check with the appropriate criteria and review period. Any trial or process change must follow the authorized procedure rather than become an improvised production experiment.

A Not here. This line belongs to turn 20, after Rosa says: "I will revise the status: containment active, supplier and setting hypotheses open, and record reconciliation assigned to you. We will not announce a cause before the evidence supports it." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Jun's response in turn 18. It connects Rosa's preceding point with Rosa's follow-up. An effectiveness check evaluates whether the action achieved the intended prevention result rather than merely being documented.

C Not here. This line belongs to turn 4, after Rosa says: "I will keep the dates and remove the confirmed-cause label. Which competing explanations do you want the investigation team to check?" It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 8, after Rosa says: "What should the evidence package contain? The identifiers in our defect, receiving, and machine records do not yet line up." It addresses that point, not the missing turn in the supplied exchange.

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Review 3 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: Work performed to restore function after a fault is identified. Which term fits?

A. corrective maintenance

A Correct. corrective maintenance: Work performed to restore function after a fault is identified. The usual combination is "plan corrective maintenance".

B Not here. external setup: Setup activity that can be completed while the machine is running. The usual combination is "separate external setup tasks".

C Not here. critical asset: Equipment whose failure has significant operational or other consequences. The usual combination is "identify critical assets".

D Not here. wear indicator: A measured or observed sign of component deterioration. The usual combination is "monitor wear indicators".

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2. A colleague needs the term for this meaning: Energy that remains after an initial control or shutdown action. Which term fits?

A. residual energy

A Correct. residual energy: Energy that remains after an initial control or shutdown action. The usual combination is "assess residual energy".

B Not here. equipment-specific procedure: Approved instructions applicable to identified equipment and work. The usual combination is "consult the equipment-specific procedure".

C Not here. personal protective equipment: Protective items worn to reduce exposure to specified hazards. The usual combination is "select required personal protective equipment".

D Not here. EHS: Environment, health, and safety functions and practices. The usual combination is "raise an EHS concern".

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Review 3 / Explanations / continued

3. Rebuild the exchange from "The supplier changed, but so did the process". Rosa: "Can you link each run to the actual material lot? The monthly supplier summary will not tell us whether the old stock was still being used." Jun: [missing response] Rosa: "The sorting team has been screening output while we investigate. The draft calls that corrective action complete, which sounds stronger than the work establishes." Which response belongs in the gap?

C. That is where lot traceability matters. Match the actual lots, settings, and observations before assuming that every unit made after the announcement used the new material.

A Not here. This line belongs to turn 2, after Rosa says: "The draft says the new supplier is the root cause because the defect first appeared after the switch. I want to check whether the evidence really isolates that change." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 4, after Rosa says: "I will keep the dates and remove the confirmed-cause label. Which competing explanations do you want the investigation team to check?" It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Jun's response in turn 10. It connects Rosa's preceding point with Rosa's follow-up. Lot traceability connects specific material lots with production and defect records instead of relying on broad period labels.

D Not here. This line belongs to turn 12, after Rosa says: "The sorting team has been screening output while we investigate. The draft calls that corrective action complete, which sounds stronger than the work establishes." It addresses that point, not the missing turn in the supplied exchange.

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Review 3 / Explanations / continued

4. Rebuild the exchange from "Stop at the unclear instruction". Hana: "I will keep the concern specific: missing equipment identity and procedure. The training card needs correction, rather than a handwritten guess about the missing steps." Oscar: [missing response] Hana: "Then this step stays on hold until the reviewed instruction is available. I'll make sure nobody mistakes our clarification for permission to start cleaning." Which response belongs in the gap?

C. I will refer it through the EHS and training process. We should preserve the unclear wording so the responsible people can review and correct the source.

A Not here. This line belongs to turn 2, after Hana says: "I want to stop at this line. The card says switch off, then clean, but it does not identify the machine or the procedure we should be using." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 8, after Hana says: "Which machine is this card supposed to cover? Can we get its approved instructions before the class goes any further?" It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Oscar's response in turn 18. It connects Hana's preceding point with Hana's follow-up. EHS identifies the environment, health, and safety function involved in addressing the safety-related training issue.

D Not here. This line belongs to turn 12, after Hana says: "Some learners also treat a tag as though it physically prevents the same actions as a lock. We should make the distinction clear without demonstrating an unapproved method." It addresses that point, not the missing turn in the supplied exchange.

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Review 4 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: The stated target or reference size. Which term fits?

C. nominal dimension

A Not here. receiving record: Documentation of material received and its identifying details. The usual combination is "trace the receiving record".

B Not here. measurement unit: The scale used to express a measurement, such as millimeters. The usual combination is "state the measurement unit".

C Correct. nominal dimension: The stated target or reference size. The usual combination is "identify the nominal dimension".

D Not here. document discrepancy: A mismatch between relevant records or references. The usual combination is "resolve a document discrepancy".

Return to the exercise, page 93

2. A colleague needs the term for this meaning: The designated review of the required evidence before a release decision. Which term fits?

C. final quality review

A Not here. shift handoff: Transfer of relevant work status and responsibility between shifts. The usual combination is "conduct a shift handoff".

B Not here. named owner: A specifically identified person responsible for an action. The usual combination is "assign a named owner".

C Correct. final quality review: The designated review of the required evidence before a release decision. The usual combination is "complete the final quality review".

D Not here. escalation trigger: A condition requiring referral to the next responsible level. The usual combination is "define an escalation trigger".

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Review 4 / Explanations / continued

3. Rebuild the exchange from "Stop at the unclear instruction". Hana: "I understand the need to ask, but I do not want my question to sound as though I am volunteering to perform the isolation myself." Oscar: [missing response] Hana: "Some learners also treat a tag as though it physically prevents the same actions as a lock. We should make the distinction clear without demonstrating an unapproved method." Which response belongs in the gap?

A. You are not. The authorized employee has an assigned role and the required training. Participating in this discussion does not give you that authorization.

A Correct. This is Oscar's response in turn 10. It connects Hana's preceding point with Hana's follow-up. The authorized employee has the assigned role and required training for the relevant energy-control work.

B Not here. This line belongs to turn 12, after Hana says: "Some learners also treat a tag as though it physically prevents the same actions as a lock. We should make the distinction clear without demonstrating an unapproved method." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Hana says: "I want to stop at this line. The card says switch off, then clean, but it does not identify the machine or the procedure we should be using." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 8, after Hana says: "Which machine is this card supposed to cover? Can we get its approved instructions before the class goes any further?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 93

Review 4 / Explanations / continued

4. Rebuild the exchange from "Six checks complete, two still open". Diego: "If a required record is unavailable, we need the team to use the site's escalation process instead of treating silence as permission to move the lot." Elena: [missing response] Diego: "The corrected record now names the lot, evidence, two open checks, dependency, release restriction, owner, and update time. That is clearer than a readiness percentage." Which response belongs in the gap?

D. I will state that escalation trigger in the handoff. The appropriate route remains the site process; I will not invent an alternative approval because a record is delayed.

A Not here. This line belongs to turn 10, after Diego says: "I will delete should be fine. What exact status would you want the incoming team to see first?" It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 2, after Diego says: "This handoff covers lot L27. Six of the eight required checks are complete, but the draft says seventy-five percent ready and should be fine." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 8, after Diego says: "Thank you. The site procedure requires all eight checks before release, so the two unfinished checks mean L27 has not been released." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Elena's response in turn 18. It connects Diego's preceding point with Diego's follow-up. The trigger identifies the missing evidence condition that requires referral through the site's responsible process.

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Additional case 1 / Explanations

Extended conversation

1. as-found results

Turn 2. As-found results describe the instrument before adjustment, which is the relevant starting point for assessing earlier measurements.

2. as-left results

Turn 4. As-left results describe the instrument after the calibration work or adjustment and do not establish its earlier condition.

3. asset identifier

Turn 6. The asset identifier connects the specific instrument to its inspection records, avoiding assumptions based on a tool description alone.

4. measurement uncertainty

Turn 10. Measurement uncertainty describes the dispersion associated with a measured value; it contributes to evaluating whether the measurement is fit for its intended decision.

5. reinspection

Turn 12. Reinspection is a new examination under an established plan; its results must remain distinguishable from the original inspection.

6. fitness for purpose

Turn 16. Fitness for purpose means the measurement capability is adequate for the intended task, not merely that a calibration label exists.

Return to the conversation, page 96

Transfer exchange

1. measurement

Supervisor: The label does not establish fitness for this measurement.

Turn 1. A calibration label alone does not show suitability for the particular measurement task.

Additional case 1 / Explanations / continued

2. limits

Engineer: Compare the instrument capability with the feature and its limits.

Turn 2. The relevant feature limits help determine the measurement capability required for the decision.

3. inspection

Supervisor: Keep earlier results separate from any new inspection.

Turn 3. New inspection results must not silently overwrite the original evidence.

4. instrument

Engineer: Record the basis for selecting the replacement instrument.

Turn 4. The selection should identify the actual instrument and why it is suitable.

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Additional case 2 / Explanations

Extended conversation

1. part-number mix

Turn 2. The part-number mix describes the quantities of different variants; the correct combined total does not establish the required mix.

2. shortage

Turn 4. The shortage is twenty right-hand brackets relative to the forty required; excess left-hand brackets do not substitute for them.

3. pick list

Turn 6. The pick list records the items and quantities requested for retrieval, helping locate where the variant mismatch entered the process.

4. kanban

Turn 8. A kanban is a signal authorizing production or movement within a defined pull system; it must identify the relevant item and quantity.

5. inventory balance

Turn 10. The inventory balance must reflect the actual quantity and location after the physical movement, rather than retaining an incorrect delivery total.

6. Held stock

Turn 14. Held stock is restricted pending the applicable decision; physical availability does not mean it is available for production.

Return to the conversation, page 100

Transfer exchange

1. assemblies

Leader: We can build only six complete assemblies.

Turn 1. Six small housings limit production to six assemblies despite the correct overall count.

Additional case 2 / Explanations / continued

2. housings

Handler: We are short of six small housings.

Turn 2. The requirement is twelve small housings but only six have arrived.

3. record

Leader: Reconcile the two variants separately on the delivery record.

Turn 3. A combined total hides the incorrect mix and must not replace variant-level quantities.

4. receipt

Handler: Verify the corrected quantities when the cell confirms receipt.

Turn 4. Receipt confirmation establishes that the corrected delivery reached the intended cell.

Return to the conversation, page 102

Additional case 3 / Explanations

Extended conversation

1. ballooned drawing

Turn 2. A ballooned drawing numbers the features to connect them with inspection results; the number must refer to the correct feature and revision.

2. inspection characteristic

Turn 4. The inspection characteristic is the specific feature or requirement being evaluated, not merely a number copied between documents.

3. controlled correction

Turn 6. A controlled correction preserves the original record and documents the authorized change, rather than silently replacing an inconsistent result.

4. datum

Turn 8. A datum is the theoretically exact reference used for applicable dimensional or geometric requirements; the reference must match the drawing and method.

5. First-piece acceptance

Turn 12. First-piece acceptance is the recorded determination that the required setup checks meet their criteria; making or submitting the piece does not establish acceptance.

6. in-process checks

Turn 14. In-process checks monitor the subsequent production according to the established plan; first-piece acceptance does not remove those requirements.

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Transfer exchange

1. accepted

Technician: The piece is submitted, but it is not yet accepted.

Turn 1. Submission is a handover milestone and does not establish the inspection decision.

Additional case 3 / Explanations / continued

2. feature

Inspector: Obtain the missing result for the specified feature.

Turn 2. The required feature result is still needed before the inspection can be resolved.

3. approval

Technician: Keep the schedule status separate from production approval.

Turn 3. A schedule label cannot substitute for the authorized production decision.

4. run

Inspector: Retain the required checks throughout the subsequent run.

Turn 4. Acceptance of a first piece does not remove the established in-process inspection requirements.

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Find a phrase by purpose

Production Flow and Daily Management

Lead with the result: We completed 840 units against a target of 1,000.

Quantify the gap: The shortfall is 160 units, or 16% of target.

Name the measure: Eighty-four percent is output target attainment.

Full phrase bank, page 7

Lean, Waste, and Continuous Improvement

Describe the delay: Operators waited for the shared tool during eight observed episodes.

Name the time basis: The total is 24 operator-minutes, not 24 minutes of line downtime.

Separate pace from access: Waiting for a tool does not by itself show slow working pace.

Full phrase bank, page 17

Defects, Scrap, and Rework

Define the population: The report covers 1,000 inspected units from this batch.

State the first-pass result: Nine hundred met requirements on the first inspection.

Separate the pending quantity: Sixty units await authorized rework.

Full phrase bank, page 30

Root Cause and Corrective Action

State the timing: The defect first appears after the supplier switch in the available records.

Identify competing changes: Two process settings changed during the same period.

Qualify the explanation: The supplier is one hypothesis, not a confirmed root cause.

Full phrase bank, page 40

Find a phrase by purpose / continued

Maintenance, Reliability, and Changeover

State the overlap: The order needs the line until 14:00, while maintenance starts at 13:00.

Keep duration distinct: A one-hour conflict does not mean a two-hour task can be completed in one hour.

Name the open assessment: We do not yet have a basis for deferring the maintenance.

Full phrase bank, page 53

EHS and Safety Communication

State the concern: Stop here: the instruction does not identify the equipment or isolation procedure.

Keep the pause clear: We should not continue from this ambiguous card.

Distinguish the actions: Switching off is not the same as verifying energy isolation.

Full phrase bank, page 64

Supplier Quality and Incoming Materials

State the result fully: The reported dimension is 12.18 millimeters.

Name the missing reference: The inspection report does not identify the drawing revision.

Show both bases: The inspector used C; the supplier used B.

Full phrase bank, page 76

Shift Handoffs and Escalation

Identify the lot: This handoff concerns lot L27.

State completed work: Six required checks are complete, with the records attached.

Name each open item: Check 7 is label reconciliation; check 8 is final quality review.

Full phrase bank, page 86

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References and publication notes

All cases, organizations, figures, and conversations are fictional teaching material, not transcripts. Communication practice does not replace qualified advice or current workplace procedures.

[1] National Institute of Standards and Technology. Value Stream Mapping.

Background on mapping material and information flow to identify improvement opportunities. The observations, pilot proposals, and datasets here are original. Source checked 10 October 2026.

https://www.nist.gov/mep/value-stream-mapping

[2] Lean Enterprise Institute. Single Minute Exchange of Die.

Background on changeover reduction and the distinction between internal and external setup. No example authorizes a machine operation. Source checked 10 October 2026.

https://www.lean.org/lexicon-terms/single-minute-exchange-of-die/

[3] American Society for Quality. Quality Glossary.

Terminology reference. Definitions and practice cases in this book are independently written for language learning, not certification or a quality-system standard. Source checked 10 October 2026.

https://asq.org/quality-resources/quality-glossary

[4] US Occupational Safety and Health Administration. Control of Hazardous Energy.

Background on the need for an appropriate energy-control program, procedures, and training. The book deliberately supplies no machine-specific isolation sequence. Source checked 10 October 2026.

https://www.osha.gov/control-hazardous-energy/

Companion course

The website offers the eight-lesson course and related practice. This book can be used independently. No audio access or external account is needed for its exercises.

https://englishladder.com/efsp-manufacturing.html

Describe the loss. Test the explanation. Make the handoff clear.

Eight shop-floor and management conversations connecting production facts to clear, accountable next steps.

Production Flow and Daily Management

840 divided by 1,000 is 84% target attainment. The shortfall is 160 units, or 16% of target. Neither result supplies OEE, availability, or first-pass yield. Those measures require their own defined inputs and calculation bases.

Lean, Waste, and Continuous Improvement

Twenty-four operator-minutes across eight episodes averages three operator-minutes per episode. Overlapping waits mean that total is not automatically elapsed line downtime. State the observation period and time basis when reporting the result.

Defects, Scrap, and Rework

900 first-pass units plus 60 awaiting rework plus 40 scrap equals 1,000 inspected units. First-pass yield is 90%; approved scrap is 4% of this population. The remaining 6% is pending rework, not already recovered output.

Root Cause and Corrective Action

After the supplier switch describes timing, not a proven causal relationship. Two settings also changed, so those variables can confound the apparent supplier effect. Reconcile comparable records before treating one explanation as established.

Phil Stilwell

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